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Volume 04 — Mobility, Cloud, Cellular, VoIP, and DDoS

Course: CEC / SWAYAM Information Security · **Coordinator:** Dr. Maninder Singh, Thapar Institute

Modules M36-M47: Mobile IP, cloud architecture and security, GSM, MANET, 3G/4G/5G, VoIP, and DDoS attacks and defenses.

These notes are grounded in the official YouTube lecture transcripts. They are study material, not an official CEC publication.

M36: Introduction to Mobile IP and Addressing

Source: <https://www.youtube.com/watch?v=I2IzxqoP7M8> **Instructor / expert:** Dr. Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana

Learning objectives

- Explain why classic IP prefixes break when a host moves, and why DHCP-only renumbering is not enough.
- Define home address vs care-of address and the roles of home agent, foreign agent, and collocated care-of address.
- Describe agent discovery (advertisement vs solicitation) and the agent-advertisement extension fields.
- Walk registration request/reply and the four-path data-transfer process (proxy ARP + tunneling).
- Distinguish double crossing (2X) from triangle (dog-leg) routing and the binding-cache fix plus its stale-cache problem.

Core concepts

Mobile communication drew a lot of attention in the last decade. Using mobiles on the Internet means the **IP protocol**, originally designed for **stationary** devices, must be enhanced for computers that **move from network to network**. The main problem is **addressing**.

Stationary hosts and why prefixes matter Original IP addressing assumed a host is **stationary**, attached to **one specific network**. A router uses an IP address to **route** a datagram. An IP address has two parts: a **prefix** and a **suffix**. The prefix **associates the host with a network**.

Example as taught: address **10.3.4.24/8** is a host on network **10.0.0.0/8**. A host therefore does **not** have an address it can **carry from place to place**. The address is valid **only while attached to that network**. If the network changes, the address is **no longer valid**.

Routers use the prefix to deliver the packet to the network the host is attached to. That scheme **works perfectly for stationary hosts**, because part of the address **defines the attached network**.

Changing the address (DHCP) — and why it fails When a host moves, the addressing structure must change. One simple proposal: let the mobile host **change its address** on the new network, using **DHCP** to obtain a new address associated with that network.

Drawbacks taught:

1. **Configuration files** would need to change.
2. Each move would require a **reboot**.
3. **DNS tables** would need revision so every other Internet host learns the change.
4. If the host **roams during a transmission**, the exchange is **interrupted**, because **ports and IP addresses of client and server must remain constant** for the duration of the connection.

Two addresses: home and care-of The feasible approach is **two addresses**:

Address	Properties
Home address	Permanent. Associates the host with its home network (permanent home)
Care-of address	Temporary. Changes as the host moves; associated with the foreign network

Mobile IP therefore gives a mobile host **one home address and one care-of address**. The home address stays; the care-of address changes from network to network.

Agents Making the address change **transparent to the rest of the Internet** requires a **home agent** and a **foreign agent**. In figures they look like **routers**, but the **agent functions are performed in the application layer** — they are **both routers and hosts**.

Home agent

- Usually a **router attached to the home network**.
- **Acts on behalf of the mobile host**.
- When a remote host sends a packet to the mobile host, the home agent **receives it and sends it to the foreign agent**.

Foreign agent

- Usually a **router attached to the foreign network**.
- **Receives and delivers** packets sent by the home agent to the mobile host.

Mobile host as its own foreign agent

The mobile host **can act as a foreign agent** (same node). To do that it must:

- Obtain a care-of address **by itself**, e.g. via **DHCP**.
- Run software to talk to the home agent and to hold **two addresses** (home and care-of) in a way that is **transparent to application programs**.

When the mobile host **is** the foreign agent, the care-of address is a **collocated care-of address**. Advantage: the host can move to **any network without needing a foreign agent to be present**.

Three phases of mobile communication To talk to a remote host, a mobile host goes through:

1. **Agent discovery** — mobile host, foreign agent, and home agent.
2. **Registration** — mobile host and the two agents.
3. **Data transfer** — the **remote host** is involved as well.

Phase 1: Agent discovery Two services:

- Discover a **home agent before leaving** the home network.
- Discover a **foreign agent after moving**, including learning the **care-of address** and the **foreign-agent address**.

Two message types: **advertisement** and **solicitation**.

Agent advertisement

When a router advertises presence with an **ICMP router advertisement**, it can **append an agent advertisement** if it acts as an agent (piggybacked on the ICMP packet).

Fields taught:

Field	Meaning
Type	8 bits, set to 16
Length	8 bits: total length of the extension , not of the ICMP advertisement
Sequence number	16 bits; recipient can detect loss
Lifetime	Seconds the agent will accept requests. All ones = infinite
Code	8-bit flags (each bit set or unset)
Care-of addresses	List of addresses usable as care-of addresses; the mobile host chooses one and announces it in the registration request . Used only by a foreign agent

Agent solicitation

If the mobile host has moved and **has not received advertisements**, it can send an **ICMP solicitation** to tell an agent it needs assistance.

Phase 2: Registration After moving and discovering the foreign agent, the host must register. Four aspects:

1. Register with the **foreign agent**.
2. Register with the **home agent** — normally done **by the foreign agent on behalf of** the mobile host.
3. **Renew** registration if it has **expired**.
4. **Cancel** registration when **returning home**.

Registration request goes mobile host → foreign agent (to register care-of address and announce home address and home-agent address). The foreign agent **relays** it to the home agent. The home agent now knows the foreign agent's address because the **relaying IP packet's source** is the foreign agent.

Request fields:

Field	Meaning
Type	8 bits; 1 for a request
Flag	8 bits of forwarding information
Lifetime	Seconds the registration is valid. All zeros = deregistration. All ones = infinite
Home address	Permanent address of the mobile host
Home agent address	Address of the home agent
Care-of address	Temporary address of the mobile host
Identification	64-bit number inserted by the mobile host and repeated in the reply to match request with reply
Extensions	Variable; used for authentication so the home agent can authenticate the mobile host

Registration reply goes home agent → foreign agent → mobile host, and **confirms or denies** the request.

Phase 3: Data transfer (four paths) Path 1 — remote host → home agent

The remote host uses **its own address as source** and the mobile host's **home address as destination**, as if the mobile host were at home. The home agent **intercepts** the packet and **pretends it is the mobile host**, using **proxy ARP**.

Path 2 — home agent → foreign agent

The home agent **tunnels**: it **encapsulates the whole IP packet** inside another IP packet, with **home-agent address as source** and **foreign-agent address as destination**.

Path 3 — foreign agent → mobile host

The foreign agent **removes** the outer packet. The inner destination is still the **home address**, so it **consults a registry table** for the **care-of address** (otherwise the packet would be sent **back to the home network**). Then it sends the packet to the care-of address.

Path 4 — mobile host → remote host

The mobile host sends **normally: home address as source**, remote host as destination. Even though the packet **leaves the foreign network**, it still carries the **home address** as source.

Transparency: the remote host is **unaware of movement**. It always sends to the home address and always receives packets sourced from the home address. **The rest of the Internet is not aware of mobility.**

Inefficiency: double crossing and triangle routing Communication involving Mobile IP can be inefficient — **severe or moderate**.

Case	Name	When	What goes wrong
Severe	Double crossing (2X)	Remote host and mobile host are on the same network	Mobile→remote is local (fine). Remote→mobile crosses the Internet twice . Because hosts usually talk to local computers, 2X waste is significant
Moderate	Triangle routing (also dog-leg routing)	Remote host is not at the same site as the mobile host	Mobile→remote is efficient. Remote→mobile goes remote → home agent → mobile: two sides of a triangle instead of one

One solution: the remote host **binds** the care-of address to the home address. When the home agent forwards the **first** packet, it can also send an **update binding** packet to the remote host so later packets go **directly to the care-of address**. The remote host keeps this in a **cache**.

Problem: the cache entry **goes stale** when the mobile host moves again. The home agent must then send a **warning packet** to the remote host.

Diagrams

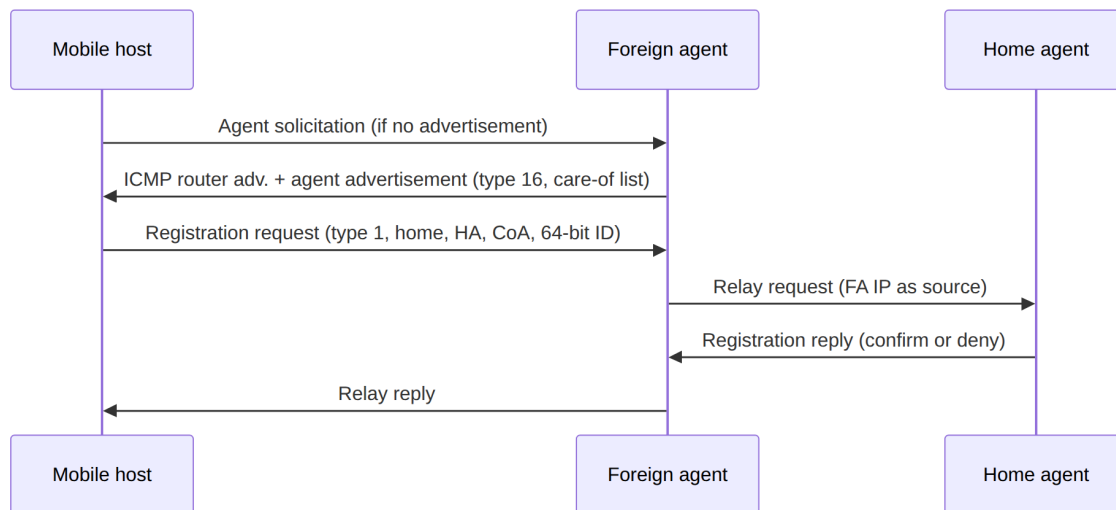


Figure 1: Architecture diagram

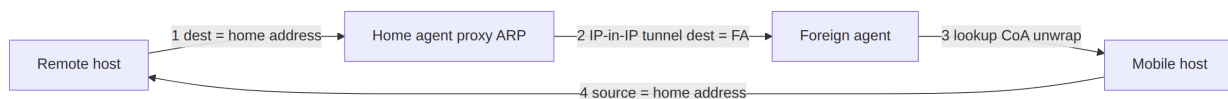


Figure 2: Architecture diagram

Key terms

Term	Meaning in this lecture
Home address	Permanent address tying the host to its home network
Care-of address	Temporary address on the foreign network
Collocated care-of address	Care-of address when the mobile host is its own foreign agent (often via DHCP)
Home agent	Application-layer agent, usually a home-network router, that intercepts and tunnels
Foreign agent	Application-layer agent on the visited network that delivers tunneled packets
Agent advertisement	ICMP router advertisement plus type-16 extension (care-of list on FA)
Agent solicitation	ICMP request for an agent when advertisements are missing
Proxy ARP	How the home agent impersonates the mobile host on the home LAN
Tunneling	Home agent encapsulates the original IP packet to the foreign agent
Double crossing (2X)	Remote and mobile on the same net; reply path still detours via home
Triangle / dog-leg routing	Remote → home agent → mobile instead of a direct path
Binding / warning	Cache CoA at the correspondent; invalidate it after a move

Lecture takeaways

- IP prefixes assume a **fixed** attachment point; DHCP renumbering breaks configs, DNS, and **in-flight connections**.
- Mobile IP keeps a **permanent home address** plus a **changing care-of address**, with HA/FA (or a collocated CoA) hiding the move.
- Discovery (ICMP advertisement type **16** / solicitation) then registration (type **1**, lifetime zeros = deregister) then four-path transfer with **proxy ARP** and **IP-in-IP**.
- Transparency is complete for the correspondent — and that is why **2X** and **triangle routing** waste path length.
- Binding caches can shortcut the triangle until the host moves; then the home agent must **warn**.

M37: Cloud Computing Architecture

Source: https://www.youtube.com/watch?v=-Ig-ll_3kAk **Instructor / expert:** Dr. Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana

Learning objectives

- Define cloud computing versus local storage and versus a “simple remote machine.”
- Recite the essential characteristics without which, the lecture says, a service is not a cloud.
- Contrast private, public, community, and hybrid deployment, including cloud bursting.
- Distinguish SaaS, PaaS, and IaaS (SPI) with the examples given (CRM, App Engine, Azure, AWS).
- Draw the four-layer architecture and state which layers fall under SLAs.

Core concepts

In simple terms, cloud computing means **storing and accessing data and programs over the Internet from a remote location or computer**, instead of the local hard drive. That remote location has properties such as **scalability and elasticity** that make it **significantly different from a simple remote machine**. The **cloud is a metaphor for the Internet**.

Storing or running a program on the **local hard drive** is **local storage and computing**. To count as cloud computing, data or programs must be accessed **over the Internet**. The end result can look the same, but with an online connection it can be done **anywhere, anytime, and by any device**.

IT and business resources — **servers, storage, network, applications, and processes** — can be **dynamically provisioned** to user needs and workload. A cloud can provision and support a **grid**, and also **non-grid** environments such as a **three-tier web architecture** running traditional or **Web 2.0** applications.

Cloud computing is a mechanism of **hiring computing power or infrastructure** at organizational or individual level **to the extent required, paying only for consumed services**. Security can be added while accessing those remote resources. Figures show several applications: the cloud is **Internet-based computing resources**, reached through **secure connectivity**.

It is growing especially among **individuals and SMEs**. It comes into focus when one asks what computing resources and IT solutions are required: a way to **increase capacity or add capabilities on the fly** without investing in new infrastructure, training new personnel, or licensing new software.

Essential characteristics The instructor stresses the word **essential**: if **any** of these is missing, **it is not cloud computing**. The spoken list overlaps NIST-style wording; all of the named properties are below.

On-demand self-service. Flexibility to **self-provision** resources according to customer demand **without being aided by any human**.

Network / broad network access. Heterogeneous network access for **mobile clients** and other computer models. Resources are reachable **from anywhere**: a **browser**, a **desktop application** designed for them, or a **mobile device**. A

popular application model cited: **iPhone apps** that communicate with a cloud-based backend.

Resource pooling (elastic / location-independent). Provider resources are **pooled** to serve multiple consumers in a **multi-tenant** model. Physical and virtual resources are **dynamically assigned and reassigned** by demand. The customer generally has **no control or knowledge of the exact location**, but may specify location at a **higher abstraction** (countries, regions). Examples of pooled resources: **storage, processing, memory, network bandwidth**. To the consumer, capacity often appears **unlimited** and purchasable in **any quantity at any time**. Resources are aggregated irrespective of consumer geolocation.

Rapid elasticity. Manage pools **on demand**: take what is needed and **free resources** when the situation changes.

Measured service. Metering **controls, monitors, and reports** resource use. Audited reports for provider and consumer show **how much disk was used**, which services were accessed, and **for how many hours**. Companies **pay only for what they use** — cost-efficient. Use can be monitored, controlled, and reported with **transparency** for both sides.

Deployment models Deployment models describe **how** cloud services are made available, depending on organizational structure and **where** data and services are provisioned. Four forms:

Model	Also called / shape	As taught
Private	Internal cloud	Platform on a cloud-based secure environment behind a firewall, under the corporate IT department. Only authorized users; the organization has direct control over data. Physical computers may be hosted internally or externally, but resources come from a distinct pool. Suited to dynamic, mission-critical needs, security, management, and uptime. Can evade some public-cloud security obstacles, but remains prone to natural disaster and internal data theft. What “counts” as private is hard to define because service mixes vary

Model	Also called / shape	As taught
Public	"True" cloud hosting	Services over a network open for public usage. Customers have no distinguished control over infrastructure location. Technically, structure may differ little from private cloud except security level. Suited to load management, SaaS-style hosted apps, and apps many users consume. Lower capital and operating cost; dealer may be free or pay-per-user license; cost shared, economies of scale. Example of a free public cloud: Google Mutually shared among organizations in a particular community — example: banks and trading firms. Members share similar privacy, performance, and security concerns and business objectives. Managed internally or by a third party; hosted externally or internally. Cost shared inside the community → cost saving
Community	Shared by a community	

Model	Also called / shape	As taught
Hybrid	Bound combination	Arrangement of two or more clouds (private, public, or community) bound together but remaining individual entities . Crosses isolation/boundaries so it cannot be simply labeled public, private, or community. Lets users increase capacity by aggregation, assimilation, or customization with another package. Resources in-house or external. Workload exchanges between private and public as needed

Hybrid examples and features

- **Non-critical** work (development and test) in a **third-party public** cloud; **critical/sensitive** work **internal**.
- **E-commerce site** on a **private** cloud for **security and scalability**; a **brochure site** (security not prime) on a cheaper **public** cloud.
- Demand spikes: extra resources from the public cloud = **cloud bursting**.
- **Big data** retained/processed with sales and business data on private cloud; **analytical queries** on the public cloud because public cloud handles spikes well.
- Hybrid hosting: **scalability, flexibility, and security**, if one overlooks challenges such as **API incompatibility, network connectivity issues**, and **capital expenditures**.

Selecting the right hosting type after analyzing business demand lets an organization channel effort into strategy. Cloud hosting has “a lot of potential,” but **selection matters**.

Three service-offering models (SPI) Cloud resources reach customers as:

Model	Lecture gloss
SaaS	Software as a Service — applications hosted by a vendor/provider and offered over a network

Model	Lecture gloss
PaaS	Platform as a Service — OS and associated services (e.g. CASE tools, IDEs) for developing solutions over the Internet without download or install
IaaS	Infrastructure as a Service — outsourcing equipment that supports operations: storage, hardware, servers, networking

Also called the **SPI** (Software / Platform / Infrastructure) model.

Cloud SaaS. Consumer uses the provider's applications running on cloud infrastructure (network, servers, OS, storage, even individual application capabilities), with possible **limited user-specific configuration**. Access from various clients via a **thin-client interface** (web browser) or a program interface. Consumer **does not manage or control** the underlying infrastructure. Typical apps: **CRM, business intelligence / analytics, online accounting**.

Cloud PaaS. Consumer **deploys** consumer-created or acquired applications built with languages, libraries, services, and tools the **provider supports**. No control of underlying infrastructure; control of **deployed applications** and possibly **hosting-environment configuration**. A **packaged, ready-to-run** development or operating framework. The PaaS side provides networks, servers, storage, and manages **scalability and maintenance**; client typically **pays for services used**. Examples: **Google App Engine** and **Microsoft Azure**.

Cloud IaaS. Consumer **provisions processing, storage, networks, and other fundamental resources** on a **pay-per-use** basis and deploys software including **OS and applications**. No control of the underlying cloud; control of **OS, storage, deployed apps**, and possibly **limited networking components**. Provider **owns** the equipment and handles **housing, cooling, operation, and maintenance**. Example: **Amazon Web Services (AWS)**.

PaaS vs IaaS: the difference is **how much control users have**. PaaS lets **vendors manage everything**; IaaS **requires more management from the customer**. If an organization **already has a software package** and wants to **install and run it on the cloud**, it should choose **IaaS instead of PaaS**.

Four-layer cloud architecture The cloud has a **hierarchical** architecture of dependencies and components. It is **completely dependent on the Internet**. Four layers based on **how the user accesses** the cloud:

Layer 1 — user / client (lowest). Where the client **initiates connections**. Device may be a **thin client, thick client, mobile**, or handheld that can access a web application.

- **Thin client:** completely dependent on another system; **very low processing**.
- **Thick client:** ordinary computer with **adequate independent processing**.

A cloud application is often accessed like a web application, but **internal properties differ**. This layer is the **client devices**.

Layer 2 — network. Lets users connect; the whole infrastructure **depends on this connection**. For a **public** cloud this is primarily the **Internet**. The public cloud exists in a specific location the user **does not know** (abstract) and can be worldwide. For a **private** cloud, connectivity may be a **LAN**, but the cloud still **depends on that network**. Users typically need a **minimum bandwidth**, sometimes defined by the provider.

This layer does **not** come under **SLA** purview. SLAs **do not** take the Internet connection between user and cloud into account for **QoS**.

Layer 3 — cloud management. Software that manages the cloud: a **cloud OS** as interface between **data center and user**, and management software for resources — **scheduling, provisioning, optimization, server consolidation, storage, workload consolidation, internal cloud governance**.

This layer **does** fall under SLAs: delays or discrepancies in provisioning can **violate the SLA** and trigger a **penalty** from the provider. SLAs apply to **private and public** clouds. Popular providers: **AWS** and **Microsoft Azure** (public); **OpenStack** for **private** cloud creation, deployment, and management.

Layer 4 — hardware resource. Actual hardware. Public cloud: a **data center** in the back end. Private: a data center (huge interconnected hardware in a location) or a **high-configuration system**. This layer is under SLAs and is the **most important layer governing SLAs**. When a user accesses the cloud it should be available **as quickly as possible** within SLA time; discrepancy means **penalty**. Data centers therefore need **high-speed network connections** and **highly efficient algorithms** to move data from the data center to the manager. A cloud may have **many data centers; several clouds can share a data center**.

There can be **loose isolation between layers 3 and 4** depending on how the cloud is deployed.

Diagrams

Key terms

Term	Meaning in this lecture
Cloud	Metaphor for Internet-based, elastic, metered remote compute — not a simple remote PC
On-demand self-service	Customer provisions without a human intermediary
Broad network access	Browser, desktop app, or mobile (e.g. iPhone apps) from anywhere
Resource pooling	Multi-tenant dynamic assignment; location opaque except region/country
Rapid elasticity	Expand and release pools as demand changes

Term	Meaning in this lecture
Measured service	Metering and audited pay-per-use reports
Private / public / community / hybrid	Four deployment models
Cloud bursting	Hybrid spike handling by drawing public-cloud capacity
SaaS / PaaS / IaaS (SPI)	App / platform / infrastructure service models
Thin vs thick client	Dependent low-power vs adequate local processing
SLA	Contract whose QoS penalties apply to management and hardware layers, not the user's Internet path
OpenStack	Example stack for private-cloud create/deploy/manage

Lecture takeaways

- Cloud is Internet access to pooled, elastic, metered resources — **hire and pay for consumption**, not a new local box.
- **All** listed essential characteristics are required, or it is **not** cloud computing.
- Choose deployment by control and community: private for mission-critical (still disaster/insider-prone), public for scale/cost (Google example), community for shared concerns (banks/traders), hybrid for bursting and mixed sensitivity.
- **SPI**: SaaS uses apps; PaaS deploys onto a vendor-managed stack (App Engine, Azure); IaaS rents raw resources (AWS). Bring-your-own package → **IaaS not PaaS**.
- Four layers from client device to data center; **network is outside SLAs, management and hardware are inside**, with hardware affecting penalties most.

M38: Security in Cloud Computing

Source: <https://www.youtube.com/watch?v=5-6WRtoGJcQ> **Instructor / expert:** Dr. Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana

Learning objectives

- Define cloud computing security as procedures, processes, and standards spanning physical and logical issues in SaaS/PaaS/IaaS and public/private/hybrid delivery.
- List data-security and virtual-data-center challenges (multi-tenancy, EU privacy, disk wipe standards, loss of visibility, provider insiders, hypervisor I/O).
- Enumerate cloud **network** challenges from guaranteed bandwidth through location-dependent VLANs.

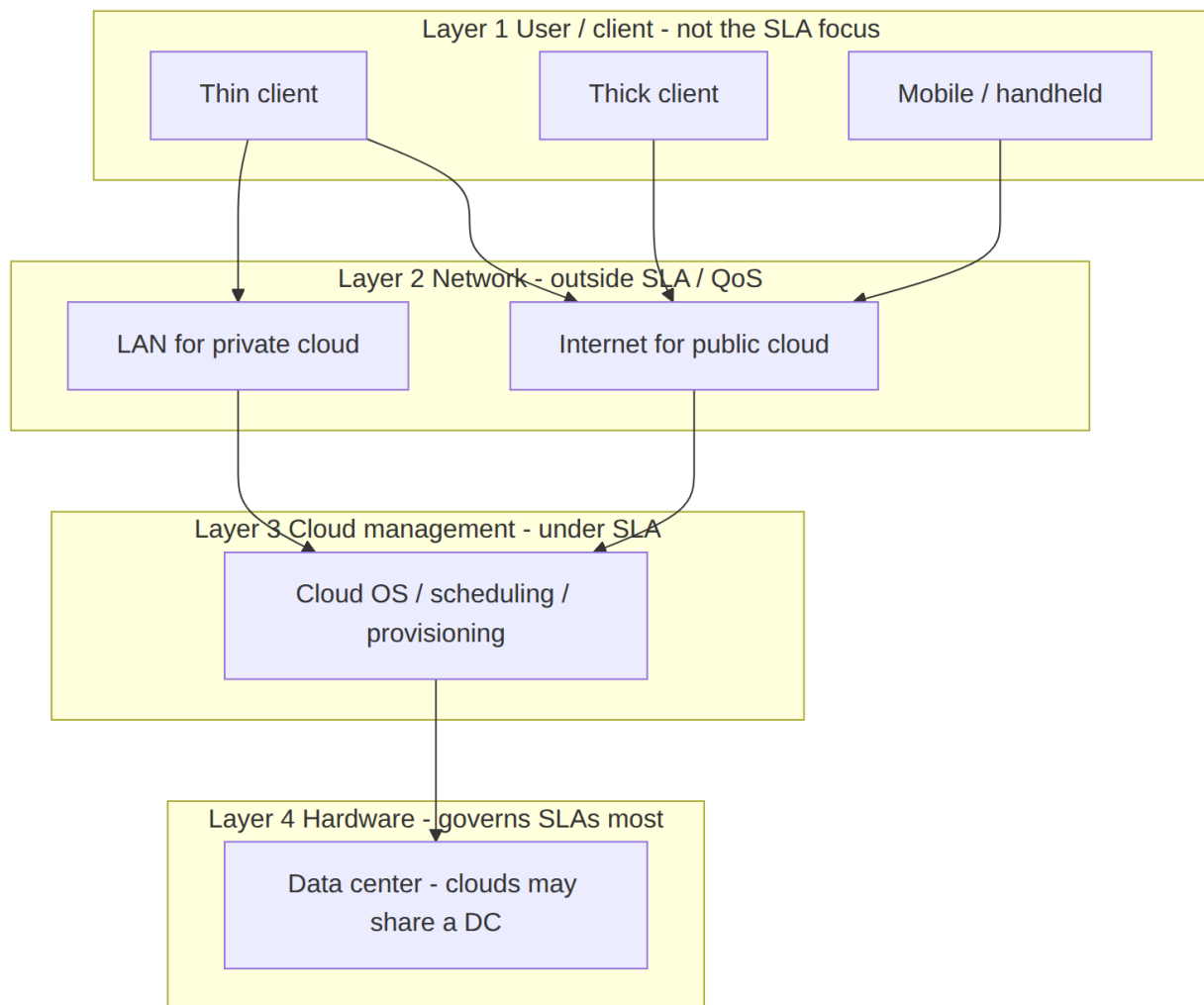


Figure 3: Architecture diagram

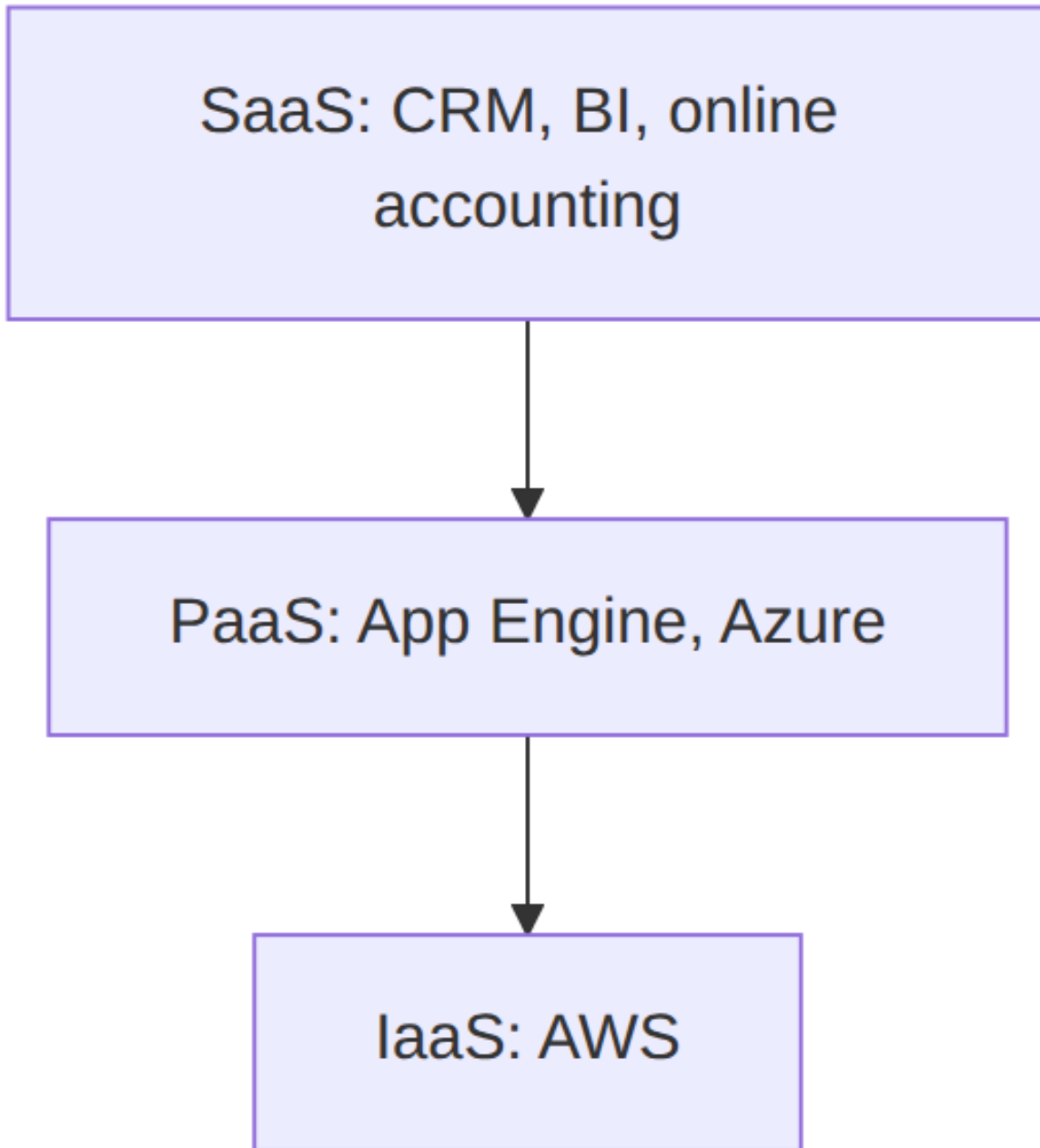


Figure 4: Architecture diagram

- Explain how security responsibility shifts as capabilities (and risks) are inherited up the SPI stack.
- Detail SaaS, PaaS, and IaaS (especially private-cloud) security issues named in the lecture.

Core concepts

Security is an important aspect of the cloud environment. **Cloud computing security** is the **set of procedures, processes, and standards** designed to provide **information-security assurance** in a cloud.

It addresses both **physical and logical** security across **software, platform, and infrastructure** service models, and how those services are delivered — **public, private, or hybrid**.

Cloud security is a broad set of constraints from **end-user and provider** perspectives. The end user is primarily concerned with the **provider's security policy: how and when data are stored and who has access**. Subsequent sections cover **data security, virtualization security**, and issues in **SaaS, IaaS, and PaaS**.

Security aspects and research context Cloud computing places business data in the hands of an **outside provider** and makes **regulatory compliance inherently riskier and more complex** than in-house systems. **Loss of direct oversight** means the client must **verify** that the provider is working to keep data security and integrity **ensured**.

Current security-related research areas taught:

- **Reliable distributed Internet applications** (e.g. **e-commerce**) that rely heavily on a **trust path** among parties.
- Skyrocketing demand for cloud consumer and business apps driving **next-generation data centers** that must be **massively scalable, efficient, agile, reliable, and secure**, so cloud services can scale to **millions of developers and billions of end users**.
- Future network-based providers will **leverage virtualization** to allocate the **right levels** of virtualized **compute, network, and storage** to each application from **real-time business demand**, with full SLA assurance of **availability, performance, and security** at reasonable cost — an evolution likened to **scalable telecommunication networks**.

Data security Because of **huge infrastructure cost**, organizations are switching to cloud. Data sit in the **provider's infrastructure, not in the organization's territory**, which raises complex challenges:

- **Service models with multiple tenants sharing the same infrastructure.**
- **Data mobility** and legal issues relative to rules such as the **European Union data privacy directive**.
- **Lack of standards** on how providers **securely recycle disk space and erase existing data**.

- **Loss of visibility** into key security and operational intelligence that no longer feeds **enterprise IT security intelligence and risk management**.
- A **new type of insider** who **does not even work for your company** but may have **control and visibility into your data**.

Data-center / virtualization security Data are stored **outside the user's territory** in a **data center whose location is unknown** — a **virtual data center**. Its backbone is **virtual infrastructure / VMs**. Virtual platforms still depend on **often forgotten** physical and virtual data-center components.

There are typically **seven areas of concern** with any major virtual-platform implementation or migration. Issues often **do not appear in staging/testing** and show up only when VMs take on the **same load as physical machines**. Two **cornerstones** of the data center are **network and storage**. Named problems:

Lack of performance and availability. Virtualization moves many **I/O tasks tuned for hardware** into software via the **hypervisor**. The virtualization translation layer translates optimized code for the software view to the **physical CPU**.

Lack of application awareness. A limitation of **hypervisor- and kernel-based** virtualization: they only virtualize the **OS**. Virtualization is **not aware of applications** on that OS; even the applications **do not realize** they are on virtual hardware above a hypervisor.

Overflowing storage. Converting physical machines to VMs helps dynamic data centers, but hard drives become **extremely large flat-file virtual disk images**, and file storage becomes **unmanageable**.

Congested storage network. Because OS virtualization is **portable**, data traversing the storage network can **increase dramatically**. Once images are portable it is **trivial** to move VM images across the network from host to host or array to array — the same reason disk files can **overrun physical storage**.

Network security in the cloud The cloud is a network of many things; the network is the **backbone**, and it has challenges:

Application performance. A tenant should specify **bandwidth** so hosted apps perform like **on-premises** deployments. Many **tiered** applications need **guaranteed bandwidth** between server instances to finish user transactions in acceptable time and meet **SLAs**. Insufficient bandwidth imposes **significant latency**.

Flexible deployment of appliances. Enterprises deploy security appliances — **deep packet inspection, intrusion detection systems, firewalls** — plus **load balancing, caching, and application acceleration**. In the cloud, the application should still **flexibly exploit** those appliances.

Policy-enforcement complexity. **Traffic isolation** and **access control** to end users are among forwarding policies that must be enforced; they **directly impact configuration of each router and switch**.

Changing requirements. Different protocols — **OSPF**, **LAG** (link aggregation), **VRRP**, and flavors of **L2 spanning-tree**, plus **vendor-specific** protocols — make it extremely hard to **build, operate, and interconnect** a cloud network **at scale**.

Application rewriting. Apps should run **out of the box** as much as possible. For **IP addresses** and **network-dependent failover**, they may need **rewrite or reconfiguration**. Two key issues: (1) **lack of a broadcast-domain abstraction** in the cloud network; (2) **cloud-assigned IP addresses** for virtual servers.

Location dependency. Appliances and servers are typically tied to a **statically configured physical network**. A server's IP is typically determined by the **VLAN or subnet** it belongs to; VLANs/subnets rest on **physical switch-port configuration**. A VM therefore **cannot be easily and smoothly migrated**. Constrained migration **decreases utilization and flexibility**. Mapping VLAN/subnet space to physical ports also **fragments the IP address pool**.

Platform-related security and inherited risk Providers offer **SaaS**, **PaaS**, and **IaaS** with varied services and challenges: **secured network**, **locality of resources**, **accessing secure data**, **data privacy**, and **backup policy**.

Cloud computing uses three delivery models to provide **infrastructure resources, application platform, and software**. They play **different levels of security requirements**.

IaaS is the basis of all cloud services, with **PaaS built upon it** and **SaaS built upon PaaS**. Capabilities are **inherited — so are information-security issues and risks**. There are trade-offs in **integrated features, complexity versus extensibility, and security**. If the provider secures only the **lower part** of the architecture, **consumers become more responsible** for implementing and managing their own security.

SaaS (again): software deployed remotely by the application/service provider, available **on demand over the Internet**. Benefits: **improved operational efficiency** and **reduced costs**. Rapidly emerging as the **dominant delivery model** for enterprise IT services.

PaaS sits one layer above IaaS and **abstracts everything up to OS, middleware, etc.** It offers developers a **complete SDLC**: planning, design, building, deployment, testing, maintenance — everything else abstracted from the developer's view.

SaaS security issues In traditional **on-premises** deployment, sensitive data stay **inside the enterprise boundary**, under its **physical, logical, and personnel** security and access-control policies.

SaaS apps are accessed through the **web**, so **web-browser security is very important**. Information-security officers must consider methods of securing SaaS: **web-services security, XML encryption, SSL**, and other options for **data protection in transit**. That means **strong encryption** and **fine-grained authorization**.

SaaS pain points:

Issue	Teaching
Network security	Sensitive data in flight must be secured against leakage via strong traffic encryption: SSL and TLS
Resource locality	Users consume services without knowing exactly where resources are . Compliance and data-privacy rules in various countries make locality of data utmost importance in much enterprise architecture
Cloud standards	Needed across organizations for interoperability, stability, and security . Example: one provider's storage services may be incompatible with another's. Providers may introduce sticky services that make migration difficult
Data access	Driven by the organization's security policies (which employees may see which data). The cloud must adhere to those policies to avoid unauthorized intrusion. SaaS must be flexible enough to incorporate org-specific policies and provide an organizational boundary inside a cloud that hosts many organizations' processes
Data breaches	Data from many users and businesses lie together ; a breach attacks all of them . The cloud is a high-value target
Backup	Vendor must regularly back up sensitive enterprise data for quick disaster recovery , with strong encryption of backups. On Amazon S3, data at rest are not encrypted by default ; users must separately encrypt data and backups so unauthorized parties cannot access or tamper with them

PaaS security issues PaaS is a **ready-to-use platform including OS on vendor-provided infrastructure**. Challenges come mainly from **spread of user objects over cloud hosts**. **Stringently allowing object access to resources and defending objects against malicious or corrupt providers** reasonably reduce risk.

Network access and service measurement raise **secure communications and access control**. Well-known practices: **object-scale enforcement of authorization** and **undeniable traceability**.

Additionally: **user privacy** must be protected in a **public shared cloud**, so proposed solutions must be **privacy-aware**. **Service continuity** concerns enterprises considering cloud adoption, so **fault-tolerant, reliable** systems are required.

IaaS security issues (especially private cloud) Cloud promises **flexibility, agility, potential cost savings**, and a competitive edge so developers can **stand up infrastructure quickly**. Private cloud solves **many** problems but is **not that good at solving security problems**. Traditional problems that remain in a **private cloud**:

Hypervisor security. Most or all services run virtualized; the hypervisor's security model **cannot be taken for granted**. **Evaluate security models and hypervisor development**.

Multi-tenancy. Even if all tenants are from the **same company**, not all may be **comfortable sharing infrastructure** with other internal users.

Identity management and access control. Traditional data centers were comfortable with a small set of authentication repositories (**Active Directory** being one of the most popular). Private cloud must handle **authentication and authorization** for the cloud infrastructure, **tenants**, and **delegation of administration** of the cloud fabric.

Network security. Many service components communicate only over **virtual network channels**. Accessing that traffic, employing **powerful access controls** (as for physical networks), and controlling **QoS** — a key **availability** issue in the **CIA** model — are major concerns.

Combined picture Combining the **three clouds** (public, private, hybrid) with the **three service-delivery models** gives a complete environment, **interlinked by connectivity devices and information-security components**. **Virtualized physical resources, virtualized infrastructure, virtualized middleware platforms, and business applications** are provided as computing services.

Providers and consumers must maintain and establish computing security at all levels of interfaces in the cloud architecture.

Diagrams

Key terms

Term	Meaning in this lecture
Cloud computing security	Procedures, processes, and standards for assurance in cloud environments
Trust path	Dependency among parties in Internet applications such as e-commerce
New insider	Provider staff who can see/control customer data without being company employees
Sticky services	Provider lock-in that makes leaving (e.g. incompatible storage APIs) hard
Hypervisor	Translation layer whose I/O and security model must not be assumed safe

Term	Meaning in this lecture
Broadcast-domain abstraction	Missing cloud-network feature that forces application rewrites
XML encryption / WS-Security / SSL / TLS	In-transit protections listed for SaaS
Amazon S3 CIA	Example where data at rest are unencrypted by default Confidentiality, integrity, availability — QoS on virtual networks hits availability
Active Directory	Example traditional authentication repository that private cloud outgrows

Lecture takeaways

- Cloud security is **physical and logical**, across **SPI** and **public/private/hybrid**, and must answer **where data live and who can see them**.
- Putting data off-premises raises **multi-tenancy**, **EU privacy**, **wipe standards**, **lost SIEM feeds**, and **provider insiders**.
- Virtual data centers hide load-time failures: hypervisor I/O, **app-unaware VMs**, huge disk images, and **portable images flooding storage networks**.
- Cloud networks struggle with **SLA bandwidth**, **appliance insertion**, **per-switch policy**, **protocol zoo**, **no broadcast domain**, **cloud IPs**, and **VLAN/port location binding**.
- **Risks inherit up the stack**; the lower the provider's security cutoff, the more the customer must implement. SaaS adds browser/transit/locality/sticky/backup issues (encrypt S3 yourself); PaaS adds object isolation and privacy; IaaS/private cloud still has **hypervisor**, **awkward internal multi-tenancy**, **IAM**, and **virtual-network CIA**.
- Security is required at **every interface** among virtualized hardware, infrastructure, platforms, and business apps.

M39: GSM Architecture

Source: <https://www.youtube.com/watch?v=OdtB17eovb4> **Instructor / expert:** Dr. Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana

Learning objectives

- Place GSM as the dominant 2G digital cellular standard (CEPT GSM → ETSI → 3GPP) and list GSM900 / DCS1800 / PCS1900 bands.

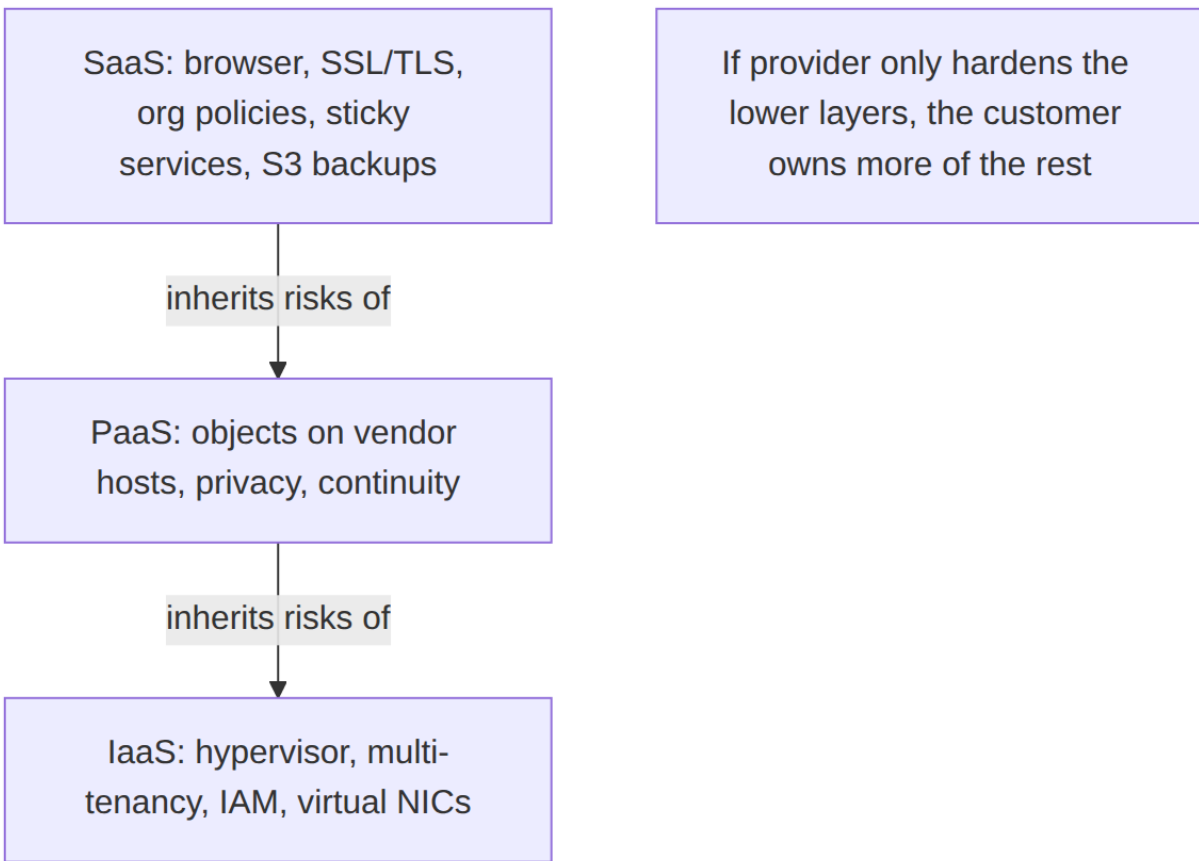


Figure 5: Architecture diagram

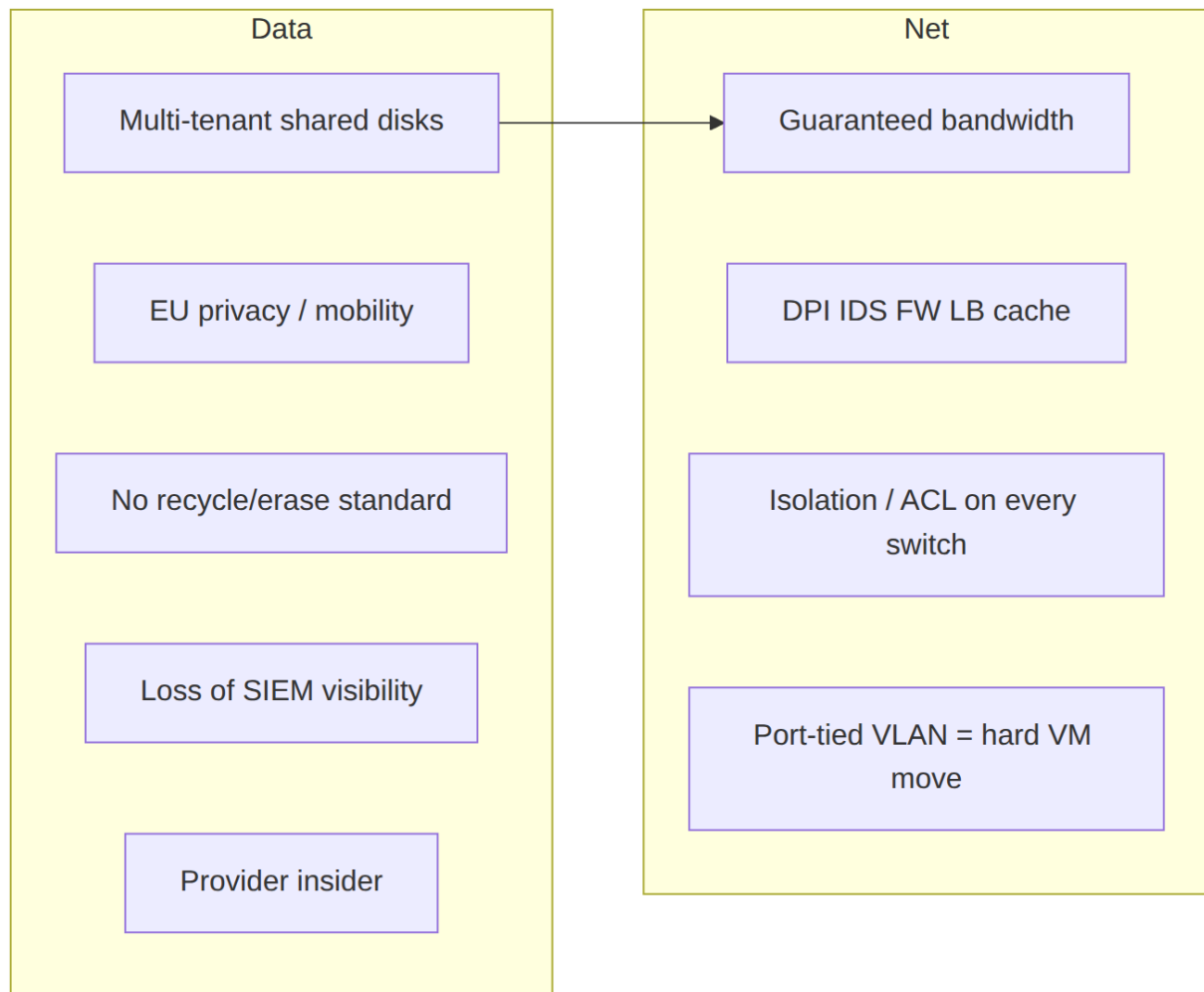


Figure 6: Architecture diagram

- Partition the network into RSS, NSS, and OSS and name the A, Abis, Um, and O interfaces.
- Identify MS hardware vs SIM identifiers (IMEI, IMSI, PIN/PUK, Ki, Kc, TMSI, LAI).
- Explain MSC, HLR, VLR, EIR, AuC, GMSC, and SMS gateways, including HLR/VLR hierarchy.
- Classify GSM teleservices, bearer services (9.6 kbps, HSCSD, GPRS), and supplementary services (CLIP/CLIR, AoC, CUG, ...).

Core concepts

GSM (Global System for Mobile Communications) is a globally accepted standard for **digital cellular** communication. Digital cellular networks are a growing market for mobile and wireless devices. They are **wireless extensions of traditional PSTN or ISDN** and allow **seamless roaming** with the same mobile phone nationally or worldwide. Today they are used mainly for **voice**, but **data traffic is continuously growing**, with several technologies for wireless data on cellular systems.

GSM is the most popular digital system, with about **70% market share**, used by **over 800 million people in more than 190 countries**.

History and bands In the early **1980s** Europe had many **coexisting analog** mobile systems, often similar standards on **slightly different carrier frequencies**. To avoid that for a **fully digital second generation**, **Groupe Spécial Mobile (GSM)** was founded in **1982**. The system was soon named **Global System for Mobile Communications**, with specification in the hands of **ETSI** (European Telecommunications Standards Institute).

In the context of **UMTS** and **3GPP** (Third Generation Partnership Project), GSM development was **transferred to 3GPP** and combined with 3G development. 3GPP **assigned new numbers** to all GSM standards.

Primary goal: a mobile phone system allowing users to **roam throughout Europe**, with **voice services compatible with ISDN and other PSTN systems**. GSM is a typical **2G** system: it **replaced 1G analog** but does **not** offer the high worldwide data rates of **3G (e.g. UMTS)**.

Name	Band as taught
GSM900	Initially Europe: 890-915 MHz uplink, 935-960 MHz downlink
DCS 1800	GSM at 1800 MHz — Digital Cellular System 1800
PCS 1900	GSM mainly used in the US at 1900 MHz — Personal Communications Service 1900

Three subsystems GSM has a **hierarchical, complex** architecture of many entities and interfaces. Three subsystems:

Subsystem	Full name	What the subscriber notices
RSS	Radio Subsystem	Mobile stations (MS) and some BTS antennas
NSS	Network and Switching Subsystem	(internal)
OSS	Operation Subsystem	(internal)

Interfaces (as drawn): RSS connects to NSS via the **A interface** (solid lines) and to OSS via the **O interface** (dashed lines).

- **A interface:** typically **circuit-switched PCM-30**, carrying up to **30 × 64 kbps** connections.
- **O interface:** **Signalling System No. 7 (SS7)** based on **X.25**, carrying **management data** to and from the radio subsystem.

Radio subsystem (RSS) RSS comprises all **radio-specific** entities: **mobile stations** and the **base station subsystem (BSS)**.

A GSM network comprises **many BSS**, each controlled by a **Base Station Controller (BSC)**. The BSS maintains **radio connections to an MS**, **coding/decoding of voice**, and **rate adaptation** to and from the wireless part. A BSS contains several **Base Transceiver Stations (BTS)**.

BTS. All radio equipment: **antennas, signal processing, amplifiers**. A BTS can form a **radio cell**, or with **sectorized antennas**, several cells. Connected to the MS via the **Um** interface and to the BSC via the **Abis** interface.

Um contains the wireless mechanisms: **TDMA, FDMA, etc.** A GSM cell can measure between some **100 m and 35 km**, depending on the environment.

BSC. Manages the BTS: **reserves radio frequencies**, handles **handover from one BTS to another within the BSS**, **pages** the MS, and **multiplexes radio channels onto the fixed network** at the **A interface**.

Mobile station (MS). The cell/mobile phone the user sees. Size has fallen while functionality has risen; time between charges has increased. Two main elements: **main hardware** and the **SIM**.

Hardware includes display, case, battery, and electronics for generating/processing the received and transmitted signal. It also contains the **International Mobile Equipment Identity (IMEI)**, installed at **manufacture** and **cannot be changed**. The network accesses it during **registration** to check whether the equipment was **reported stolen**.

The **SIM (Subscriber Identity Module)** identifies the **user** to the network. It holds identifiers and tables: **card type, serial number, list of subscribed services, PIN, PIN Unblocking Key (PUK), authentication key Ki**, and **IMSI (International Mobile Subscriber Identity)**.

- **PIN** unlocks the MS. **Wrong PIN three times locks the SIM**; then **PUK** is needed.
- While logged on, the MS stores **dynamic** information: cipher key **Kc**, and location information — **TMSI** (Temporary Mobile Subscriber Identity) and **LAI** (Location Area Identification).

Typical transmit power: up to **2 W** for **GSM900**; **1 W** is enough for **GSM1800** because of **smaller cell size**.

Besides the telephone interface, an MS can offer display, loudspeaker, microphone, programmable soft keys, computer/modems, or **Bluetooth**. Vendor-specific extras: cameras, fingerprint sensors, calendars, address books, games, Internet browsers. **PDAs** with mobile-phone functions exist. An MS can also be **integrated into a car** or used for **location tracking of a container**.

Network and Switching Subsystem (NSS) The **heart** of GSM. NSS **connects the wireless network with standard public networks**, performs **handovers between different BSS, worldwide localization**, and supports **charging, accounting, and roaming** between providers in different countries. It consists of switches and databases:

MSC (Mobile Switching Center). Main element of the core network. Acts like a normal **PSTN/ISDN switching node** plus mobility functions. Sets up connections to other MSCs and to BSCs via the **A interface**; forms the **fixed backbone**. An MSC manages **several BSCs** in a geographical region: **registration, authentication, call location, inter-MSC handovers, call routing** to a mobile subscriber. Interface to **PSTN** (landline), to other MSCs (other networks), and to **public data networks (PDN)** such as **X.25**. Handles signaling for **connection setup, release, and handover to other MSCs**.

HLR (Home Location Register). **Most important database**. Stores all user-relevant information:

- **Static:** **MSISDN** (Mobile Subscriber ISDN number), subscribed services (e.g. **call forwarding, roaming restrictions, GPRS**), **IMSI**.
- **Dynamic:** current **location area** of the MS, **MSRN** (Mobile Station Roaming Number), current **VLR and MSC**.

When the MS **leaves its current LA**, HLR is **updated**. At switch-on the phone **registers**, so the network knows which **BTS** it uses and can **route incoming calls**. Even when not in a call but **switched on**, it **re-registers periodically** so the HLR has the latest position — necessary to **localize a user anywhere** in GSM. Each user's data exist **once**, in a **single HLR**, which also supports **charging and accounting**. HLRs can manage **several million** customers and are **highly specialized real-time databases**. **One HLR per network**, though it may be **distributed across subcenters**.

VLR (Visitor Location Register). Associated with each MSC; a **dynamic** database of users currently in that MSC's location area (e.g. **IMSI, MSISDN, HLR address**). When a new MS enters an LA, the VLR **copies relevant information from the HLR**. This **HLR/VLR hierarchy avoids frequent HLR updates and long-distance**

signaling. The VLR can be separate but is **commonly integral to the MSC** for faster access.

EIR (Equipment Identity Register). Decides whether given **mobile equipment** may enter the network, using **IMEI**, checked at registration. (Also described under OSS.)

AuC (Authentication Center). **Protected database** containing the **secret key also in the user's SIM**; used for **authentication and ciphering on the radio channel**. (Also under OSS.)

GMSC (Gateway MSC). Point to which a **mobile-terminating call is initially routed without knowledge of MS location**. Obtains the **MSRN from the HLR** based on **MSISDN** (directory number) and routes the call to the **correct visited MSC**. The "MSC" in GMSC is **misleading**: gateway operation does **not** require linking to an MSC.

SMS gateway (SMSG). Collective name for two short-message gateways:

Gateway	Direction
SMS-GMSC	Short messages to a mobile (role similar to GMSC)
SMS-IWMSC (interworking MSC)	Short messages originated by a mobile on that network; fixed access point to the SMS Center

Operation Subsystem (OSS) Necessary functions for **network operation and maintenance**. OSS has its own entities and accesses others via **SS7**. Connected to components of **NSS and the BSC**. Used to **control and monitor** the overall GSM network and the **traffic load of the BSS**. As BTS count scales with subscribers, some maintenance tasks move to the **BTS** to save cost of ownership.

OMC (Operation and Maintenance Center). Monitors and controls all other entities via the **O interface**. Typical functions: **traffic monitoring, status reports, subscriber and security management, accounting and billing**. Uses the **Telecommunications Management Network (TMN)** concept standardized by the ITU.

AuC (again, OSS view). Because the radio interface and MSs are particularly vulnerable, a separate AuC protects **user identity and data transmission**. It contains **authentication algorithms and encryption keys**, and **generates values needed for user authentication in the HLR**. It may sit in a **special protected part of the HLR**.

EIR (again, OSS view). Database of all **IMEIs** registered for the network. Mobiles can be **easily stolen**; with a **valid SIM**, anyone could use a stolen MS. EIR has a **blacklist** of stolen or locked devices. **Blacklists of different providers are not usually synchronized**, so **illegal use in another operator's network is possible**. EIR also has a **whitelist** of valid IMEIs and a **grey list** of **malfunctioning** devices.

GSM services GSM offers more than voice telephony; ask the local operator which services are available. Three basic types:

Teleservices (T services). Use bearer-service abilities to transport data.

- **Voice calls:** most basic T service — **telephony**, including **full-rate speech at 13 kbps** and **emergency calls** (nearest emergency provider via **three digits**).
- **Videotex and facsimile:** videotext access, **teletex**, fax, alternate speech and automatic faxing.
- **Short text messages (SMS):** send/receive text on the GSM phone; also news, sports, financial, language, and **location-based** data.

Bearer services (data services). Used through a GSM phone to send/receive data — the building block toward **mobile Internet** and mobile data transfer. GSM currently **9.6 kbps**. Newer developments: **HSCSD** (High-Speed Circuit-Switched Data) and **GPRS** (General Packet Radio Service), now available.

Supplementary services. Additional to T and bearer services: **caller identification, call forwarding, call waiting, multi-party conversations, barring of international outgoing calls**, among others.

Supplementary service	Meaning as taught
Conferencing	Multi-party conversation (three or more); only for normal telephony
Call waiting	Notify of an incoming call during a conversation; answer, reject, or ignore
Call hold	Put an incoming call on hold and resume; normal telephony
Call forwarding	Divert from original recipient to another number; usually set by the subscriber, e.g. when not available
Call barring	Restrict certain outgoing calls (e.g. ISD) or stop incoming from undesired numbers; flexible conditional barring
CLIP	Calling Line Identification Presentation — show caller's number
CLIR	Calling Line Identification Restriction — caller hides number
COLP	Connected Line Identification Presentation — calling party sees the number actually connected (useful when forwarded)
COLR	Connected Line Identification Restriction — called party hides number; normally overrides presentation

Supplementary service	Meaning as taught
Malicious call identification	Combat obscene or annoying calls; subscriber causes unknown malicious calls to be identified in the GSM network with a simple command
AoC (Advice of Charge)	Indicate cost of services used; rental/providers without the user's own SIM can use a slightly different form; AoC for data calls is time-based
CUG (Closed User Group)	Groups who wish to call only each other and no one else

Diagrams

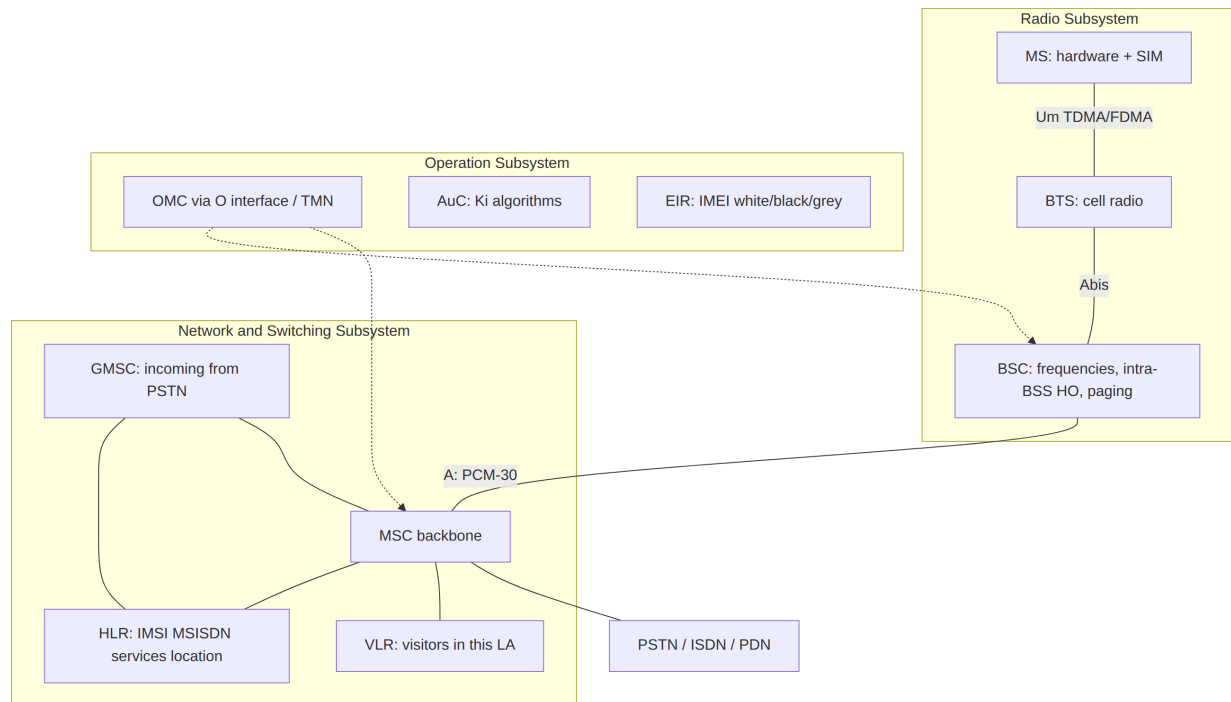


Figure 7: Architecture diagram

Key terms

Term	Meaning in this lecture
GSM	2G digital cellular; ~70% share; 800+ million users in 190+ countries
RSS / NSS / OSS	Radio, switching, and operation subsystems

Term	Meaning in this lecture
Um / Abis / A / O	MS-BTS radio; BTS-BSC; BSS-MSC (PCM-30); management (SS7/X.25)
IMEI / IMSI	Equipment identity (in hardware) vs subscriber identity (on SIM)
PIN / PUK / Ki / Kc	Unlock SIM; unblock after 3 PIN fails; auth key; cipher key
TMSI / LAI	Temporary subscriber ID and location-area ID stored while camped
HLR / VLR	Permanent user DB vs per-MSC visitor copy (cuts long-distance signaling)
AuC / EIR	Auth/cipher secrets; IMEI white/black/grey lists (blacklists not shared across operators)
GMSC / SMS-GMSC / SMS-IWMSC	Incoming voice routing; SMS toward MS; SMS from MS
Teleservice / bearer / supplementary	Speech/SMS/fax; 9.6 kbps data path (HSCSD, GPRS); extras like CLIP and CUG
MSISDN / MSRN	Directory number vs routing number obtained from HLR

Lecture takeaways

- GSM is Europe's answer to fragmented analog 1G: **one digital 2G** under ETSI then **3GPP**, built to **roam** with **ISDN-like voice**, later band-extended as GSM900 / DCS1800 / PCS1900.
- Users only see **MS + BTS antennas**; the real machine is **RSS (Um/Abis) + NSS (A, HLR/VLR/MSC) + OSS (O, TMN)**.
- Identity splits: **IMEI** (stolen phone) vs **IMSI/SIM** (stolen credentials); **VLR copies HLR** so the home database is not hit on every local event.
- Incoming calls go **GMSC → HLR (MSRN) → visited MSC**; SMS has analogous **GMSC/IWMSC** pair.
- Services stack as **teleservices (13 kbps speech, SMS, fax), bearers (9.6 kbps, then HSCSD/GPRS)**, and a long list of **supplementary** controls (CLIP/CLIR, barring, CUGs, AoC).
- EIR blacklists **do not sync between operators**, so a stolen handset may still work on another network.

M40: MANET - I

Source: https://www.youtube.com/watch?v=7CR1_FBxJ6o **Instructor / expert:** Dr Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana **Course coordinator:** Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

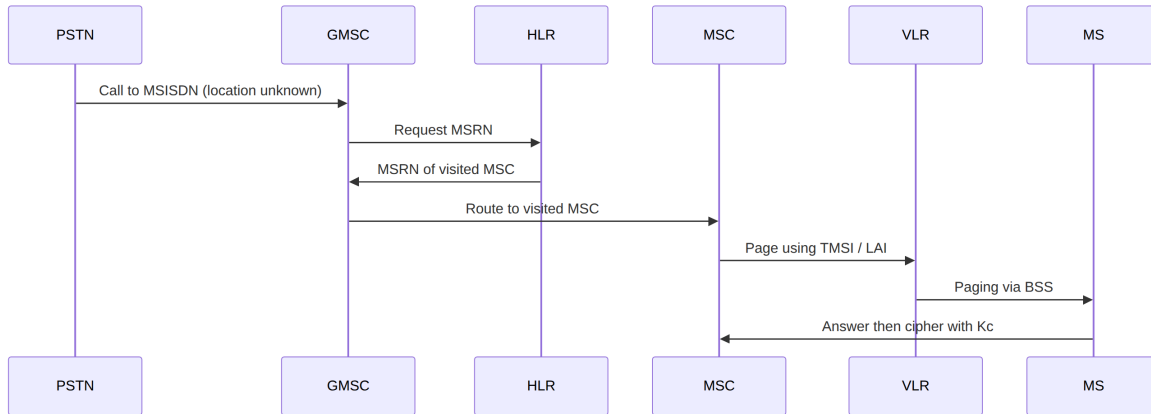


Figure 8: Architecture diagram

Learning objectives

- Contrast wired networks with wireless networks and name the three wireless types the lecture uses: fixed wireless, wireless with fixed access points, and MANET.
- Define a mobile ad hoc network as an infrastructure-less, IP-based radio network in which nodes act as both hosts and routers.
- List MANET characteristics (autonomous terminals, distributed operation, multi-hop routing, dynamic topology, limited resources and security, scalability, low bandwidth).
- State the open challenges the instructor listed: scalability, routing, QoS, the client-server model shift, security, energy conservation, node cooperation, and interoperation.
- Recall the application families taught: military, home/office/education, VANET, wireless sensor networks, wireless mesh networks, disaster relief, and personal area networks.

Core concepts

Like traditional **wired** networks, wireless networks are formed by **routers** and **hosts**. Routers forward packets; hosts are sources or **sinks** of data flows. The fundamental difference is how components communicate. A wired network relies on physical cables. In a wireless network, communication between components can be either wired or wireless. Because wireless communication is not constrained by cables, hosts and routers have **freedom to move** — one of the advantages of wireless networking.

Three types of wireless network According to the mobility of hosts and routers, the lecture distinguishes three types:

Type	How it is formed	Example given
Fixed wireless network	Fixed hosts and routers use wireless channels to communicate	Fixed devices using directed antennas ; two computers exchanging data via antenna signals
Wireless network with fixed access points	Mobile hosts use wireless channels to talk to fixed access points , which may act as routers	Laptop users in a building accessing fixed APs
Mobile ad hoc network (MANET)	Formed by mobile hosts , some of which forward packets for neighbours	Vehicle-to-vehicle and ship-to-ship networks using peer-to-peer routing

In the ship example, all ships create a network **without any centralized administration**; a node can establish a link with any other node in range.

What a MANET is A MANET is an **infrastructure-less, IP-based** network of mobile wireless machines connected by **radio**. Nodes have **no centralized administration**. It is a **routable** network: **each node acts as a router** and forwards traffic toward a specific destination. Mobiles and laptops can form such a network among themselves.

In the simplest case, nodes communicate **directly** when they are within wireless transmission range. Ad hoc networks must also support communication between nodes that are only **indirectly** connected by a series of wireless **hops** through other nodes. In the lecture's figure, nodes **A** and **C** are not in each other's radio circles, so the network must enlist **node B** to relay packets. The circles are the **nominal range of each node's radio transceiver**.

Characteristics of MANETs Compared with wired or infrastructure-based wireless networks, MANETs have the following characteristics (preview list plus the expanded treatment):

Autonomous terminal. Each mobile terminal is an autonomous node that may function as a **host or a router**. Besides host processing, nodes perform switching. End-points and switches are usually **indistinguishable**.

Distributed operation. There is no background network for central control. Control and management are **distributed among the terminals**. Nodes must **collaborate**; each acts as a **relay** as needed to implement functions such as **security and routing**.

Multi-hop routing. Ad hoc routing can be **single-hop or multi-hop**, based on link-layer attributes and routing protocols. Single-hop MANETs are simpler in structure and implementation. When the destination is **outside direct wireless range**, intermediate nodes must forward packets.

Lightweight terminals. Nodes are typically mobile devices with **less CPU, small memory, and low power storage**. They need **optimized algorithms** for computing and communicating.

Dynamic topology. All nodes are free to move, so topology changes **rapidly at unpredictable times**. Links break **much more frequently** than in wired or infrastructure wireless networks.

Self-organization. With no infrastructure or central administration, nodes must **form themselves into a network**.

Multi-hopping. The wireless channel and limited neighbourhood mean **intermediate nodes relay packets**.

Resource conservation. Nodes are limited in **energy supply and processing power**. Power conservation is a major design factor; operations should be optimized to **minimize energy consumption**.

Limited capacity / limited security. Mobile ad hoc networks are **more prone to security threats** than wired or infrastructure wireless networks. Each node can function as a router or packet forwarder. **Both legitimate users and malicious attackers** can access the wireless channel, and there is **no well-placed location at which access-control mechanisms can be deployed**.

Scalability. Some applications may grow to **several thousand nodes**. MANETs suffer scalability problems in **channel capacity**: capacities are limited, and the maximum is reached faster because of **multi-hopping**. Scalability is **related to the routing protocols** employed.

Low bandwidth. These networks have **lower capacity and shorter transmission range** than fixed-infrastructure networks. Wireless throughput is less than wired throughput because of **multipath fading, noise, and interference**.

Challenges of MANETs The instructor listed topics that still need to be resolved: **scalability, routing, quality of service, client-server model shift, security, energy conservation, node cooperation, and interoperation**. The lecture then developed several of these.

Scalability. Visionaries often take large-scale “universal computing” as granted, but it is **unclear how such large networks can actually grow**. Ad hoc networks suffer **by nature** from capacity scalability problems.

Routing. Routing is non-trivial because of a **highly dynamic** environment. An ad hoc network is a collection of wireless mobile nodes **dynamically forming a temporary network** without pre-existing infrastructure or centralized administration. Nodes come together for a period to exchange information **while continuing to move**, so the network must **continually adapt and establish routes among themselves without outside support**.

Quality of service (QoS). Heterogeneous Internet applications challenged designers who built networks for **best-effort** service only. **Voice, live video, and file**

transfer have vast and diverse requirements. **QoS-aware** solutions are being developed. QoS means the network **guarantees certain performance** for a flow or collection of flows in parameters such as **delay, jitter, bandwidth, and packet-loss probability**. Despite research, QoS in ad hoc networks is still described as an **unexplored area**.

Client-server model shift. On the Internet a client is typically configured to use a **server** (found automatically or by static configuration). In ad hoc networks the structure **cannot be defined by collecting IP addresses into subnets**. There may be **no servers**, yet demand remains for basic services: **address allocation, name resolution, authentication, and service location**. Their location is **unknown and changes dynamically**. A **different addressing approach** may be required, and it is still not clear **who manages** various network services. Shifting away from the traditional client-server model **remains to be appropriately addressed**.

Security. Applications such as **military** and **confidential meetings** need a high degree of security against enemies and **active and passive attackers**. Ad hoc networks are **particularly prone to malicious behaviour**. Lack of centralized network management or a **certification authority** in these dynamically changing wireless structures is framed as a **major roadblock to commercial application**.

Interoperation. When two autonomous ad hoc networks move into the **same area**, interference is unavoidable and the networks would need to **recognize the situation and merge**. Joining two networks is **not trivial**: they may use different **synchronization**, or even different **MAC or routing protocols**. Security is a major concern. Example: a **military unit** moving into an area covered by a **sensor network** — the unit might use a routing protocol with **location information**, while the sensor network uses a **simple static routing protocol**.

Energy conservation and **node cooperation** were named on the challenge list alongside the topics above.

Applications MANETs are best where infrastructure is **unavailable** or **not cost-effective**.

Military. Satisfies needs such as **battlefield survivability**. Setting up infrastructure among soldiers may be impossible. Wireless devices carried by **soldiers, tanks, helicopters, and other vehicles** form a MANET for message passing.

Home, office, and education. The simplest application is networking **laptops, PDAs, and other WLAN-enabled devices** in the **absence of a wireless base station**. A **personal area network (PAN)** class example is **wire replacement**: in **Bluetooth**, peripheral devices connect to a computer over wireless Bluetooth links. Ad hoc networks can **stream video and audio** among wireless nodes without a base station — e.g. students obtaining class material from a professor's laptop as class progresses. Other educational uses: **universities and campus settings, virtual classrooms, ad hoc communications during meetings or lectures**.

Vehicular ad hoc networks (VANETs). Responsible for communication between moving vehicles. A vehicle may talk to another vehicle directly (**vehicle-to-**

vehicle, V2V) or to infrastructure such as a **roadside unit (RSU) (vehicle-to-infrastructure, V2I)**.

Wireless sensor networks (WSNs). Treated as a **special case of MANET with reduced or no mobility**. Initially motivated by **military** applications; civilian domains include **environmental and space monitoring, disaster management, smart home, production, and healthcare**. WSNs may consist of **heterogeneous and mobile** sensor nodes. Topology may be as simple as a **star**. Scale and density vary by application. Figured applications: disaster management, network management, home networks, healthcare informatics, and military.

A sensor network is many **tiny, disposable, low-power devices (nodes)** spatially distributed to perform an **application-oriented global task**. Nodes communicate directly or through other nodes. One or more nodes serve as a **sink**, able to communicate with the user directly or through existing **wired** networks. The primary component is the **sensor**, monitoring real-world conditions such as **sound, temperature, humidity, vibration, pressure, motion, pollutants**, and so on at different locations. Nodes **sense, process, and communicate** in a **self-organizing**, battery-powered network.

Wireless mesh networks (WMNs). Can connect entire cities using inexpensive existing technology. Traditional networks rely on a small number of **wired access points or wireless hotspots**. In a mesh, the connection is spread among **dozens or hundreds of mesh nodes** that talk to each other to share connectivity across a large area. Mesh nodes are small radio transmitters that function like **wireless routers**. They use common **Wi-Fi** standards **IEEE 802.11a, 802.11b, and 802.11g**, on **2.4 GHz and 5 GHz** depending on the physical layer. If **IEEE 802.11a** is used, speed can be up to **54 Mbps**. Nodes are programmed with software that tells them how to interact with the large network. Information hops from node to node; nodes automatically choose the **quickest and safest path** using **dynamic routing**. The lecture figure: an **Internet cloud** as backbone of mesh routers, connected with access points, base stations, and other devices that provide Internet access to users.

Disaster relief. Earthquakes and floods destroy lives; loss of Internet connectivity is an increasingly noticeable effect. It is important to **keep networks operating when infrastructure elements are disabled**.

Personal area networks (PANs). A computer network for data among **computers, telephones, and PDAs**, consisting of nodes associated with a **single person** (clothes, belt, handbags). Flexibility examples: send a document to a printer upstairs while sitting on a couch with a laptop; upload a photo from a cell phone to a desktop.

The instructor closed by pointing to the **next lecture on MANET routing protocols**.

Key terms

Term	Meaning in this lecture
MANET	Infrastructure-less, IP-based radio network of mobile nodes that route for one another

Term	Meaning in this lecture
Access point	Fixed infrastructure node that mobile hosts associate with in the non-ad-hoc wireless type
Multi-hop	Relaying through intermediate nodes when source and destination are not in direct range
Dynamic topology	Unpredictable, rapid link change because nodes move freely
QoS	Guaranteed performance for a flow: delay, jitter, bandwidth, packet-loss probability
VANET	Vehicular ad hoc network; V2V between vehicles, V2I via roadside units
WSN	Sensor-node network treated as a low-mobility special case of MANET, with sink nodes
Wireless mesh	Many 802.11a/b/g mesh routers sharing an Internet connection by dynamic hop-by-hop routing
PAN	Person-centred short-range network, including Bluetooth wire-replacement

Lecture takeaways

- Wireless freedom of movement yields three classes; MANET is the infrastructure-less extreme.
- Every MANET node is potentially a host **and** a router; A-C communication may require B.
- Design is dominated by dynamics, scarce energy/CPU/bandwidth, and the absence of a place to hang access control.
- Routing, QoS, naming/addressing without servers, security without a CA, and merging dissimilar networks are still open.
- The same idea covers battlefields, classrooms, vehicles, sensors, city-scale meshes, disasters, and PANs.

M41: MANET - Routing Protocols

Source: <https://www.youtube.com/watch?v=9eNSUh66Zag> **Instructor / expert:** Dr Sunil Kumar, School of Electrical Engineering and Information Technology, Punjab Agricultural University, Ludhiana **Course coordinator:** Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Define what a routing protocol does in a MANET and why metrics and information exchange matter under changing conditions.
- Distinguish **reactive (on-demand)**, **proactive (table-driven)**, and **hybrid** routing, including the bandwidth and latency trade-offs the lecture stated.

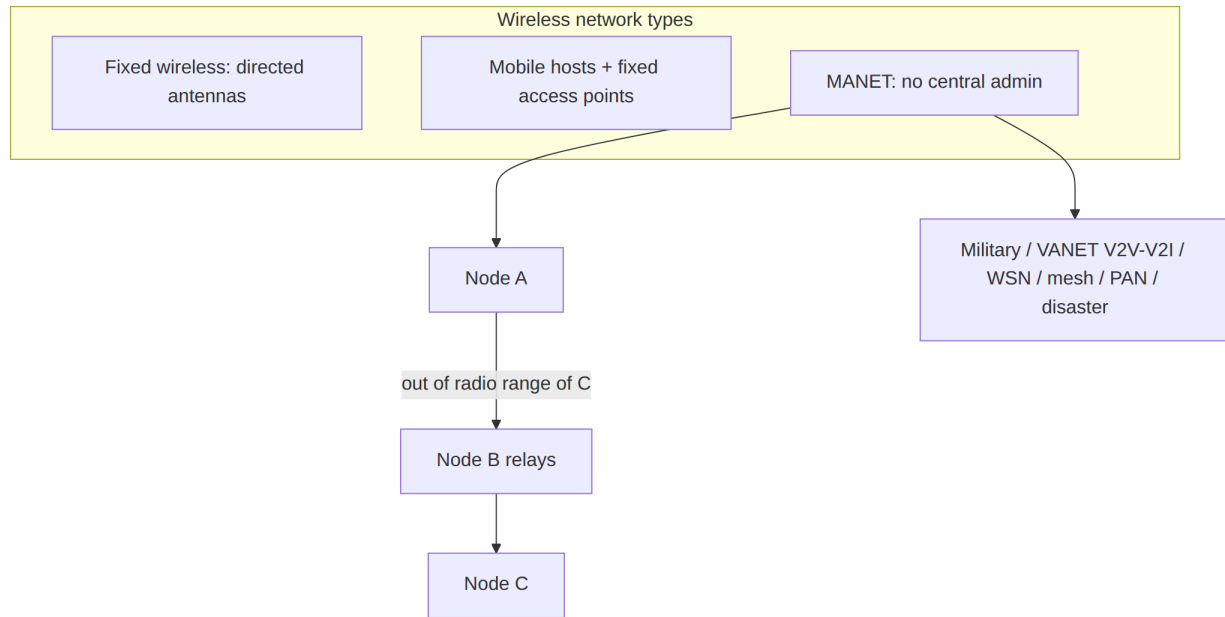


Figure 9: Architecture diagram

- Explain **AODV** (RFC 3561): RREQ / RREP / RERR, expanding-ring TTL, sequence numbers, precursor lists, and the three RERR circumstances.
- Explain **OLSR** as a proactive protocol that uses **multipoint relays (MPRs)**, Hello-based neighbour sensing, MPR flooding, topology-control (TC) messages, and Dijkstra shortest paths.
- Explain **TORA** as a hybrid protocol that builds a **DAG** of **heights** using QUERY, UPDATE, and CLEAR, with route creation, five maintenance cases, and route deletion.

Core concepts

A **routing protocol** is the relationship or formula routers use to find a suitable path for forwarding data. It also supports **exchange of information between routers** and helps the network **adjust to variable conditions**. In other words, it is the **implementation of a routing algorithm in software or hardware**. Protocols use **different metrics** to choose the path that transfers a packet across the network.

Three types of MANET routing protocol **Reactive routing protocols** are **on-demand**. They are invoked when needed, and routes are built then. Routes can be acquired by sending a **route request** through the network. The stated disadvantage is **high latency** in searching the network.

Proactive routing protocols build and maintain routing information about all nodes, independent of whether a route is currently needed. Control messages are transmitted periodically even if there is no data flow, so they are not bandwidth-efficient. Advantage: nodes can **easily get routing information and**

start a session. Disadvantages: nodes keep **too much data for route maintenance**, and **recovery after a particular link failure is too slow**.

Hybrid routing protocols combine benefits of both. Routing **initiates using proactive pre-planned routes**; extra demand is then handled by **reactive flooding**. This methodology is suitable only in situations where **traffic demand and the number of nodes can be determined beforehand**.

The lecture then developed three named protocols: **AODV** (reactive), **OLSR** (proactive), and **TORA** (hybrid).

AODV — Ad hoc On-Demand Distance Vector AODV is a **reactive** protocol that **sets up routes on demand**. If a node wants to communicate with a node for which it has **no route**, the protocol tries to establish one. It is described in **RFC 3561**.

As with all reactive protocols, **topology information is transmitted by nodes only on demand**. When a node wishes to send traffic to a host with no route, it generates a **route request (RREQ)** that is **flooded in a limited way**. Control-traffic overhead is therefore **dynamic**, and there is an **initial delay** when starting communication.

A route is considered **found** when the RREQ reaches **either the destination itself** or an **intermediate node with a valid route entry** for the destination. For as long as a route exists between two endpoints it is used; if the node remains passive when the route becomes **invalid or lost**, AODV **issues a request again**.

AODV defines **three control messages** for route maintenance:

RREQ (route request). Transmitted by a node that needs a route. As an optimization, AODV uses an **expanding-ring** technique when flooding. Every RREQ carries a **time-to-live (TTL)** stating **how many hops** the message should be forwarded. TTL starts at a **predefined value** on first transmission and is **increased on retransmissions**. Retransmissions occur if **no replies** are received.

RREQ frame fields taught: **source address, source sequence, broadcast ID, destination address, destination sequence, hop count**.

RREP (route reply). **Unicast** back to the originator of the RREQ if the receiver **is the requested address** or **has a valid route** to it. Unicast is possible because **every node forwarding an RREQ caches a route back to the originator**.

RREP fields taught: **source address, destination address, destination sequence, hop count, lifetime** (how long a path can be used).

RERR (route error). Nodes monitor the **link status of next hops in active routes**. When **link breakage** on an active route is detected, a RERR notifies other nodes of the loss. To enable this, each node keeps a **precursor list**: IP addresses of neighbours **likely to use it as next hop** toward each destination.

Worked example (four nodes). Circles are communication range; each node can talk only to neighbours. Node **1** wishes to send to node **3**. Neighbours of 1 are **2 and 4**. Node 1 **cannot** reach 3 directly, so it sends an RREQ, heard by 4 and 2. A node that receives RREQ either (a) **knows a route to the destination or is the destination** and sends an RREP, or (b) **rebroadcasts** RREQ to its neighbours.

Rebroadcast continues until the **lifespan** is up. If node 1 gets **no reply in a set time**, it rebroadcasts with a **longer lifespan** and a **new ID**. **Sequence numbers** in the RREQ ensure nodes **do not rebroadcast the same request**. In the example, **node 2 has a route to 3 and replies; node 4 does not**, so it rebroadcasts. When a node sees an RREP whose route has a **better sequence number** than its routing list, it **replaces** its current route with the one in the RREP.

When RERR is broadcast (three circumstances):

1. The node receives a **data packet it is supposed to forward** but has **no route** to the destination — the real problem is that **some other node thinks the correct route is through this node**.
2. The node receives a RERR that **invalidates at least one of its routes**; it then sends a RERR naming the **newly unreachable** destinations.
3. The node **detects it cannot communicate with a neighbour**; it marks routing-table entries that use that neighbour as next hop **invalid**, then sends a RERR with the neighbour and the invalid routes.

Whenever a node receives a RERR it **looks at the routing table and removes all routes that contain the bad nodes**.

OLSR — Optimized Link State Routing OLSR is a **proactive, table-driven** protocol: it **permanently stores and updates** its routing table so a route can be provided if needed. It can be implemented in **any ad hoc network**. Nodes collaborate by **periodically exchanging topology information**.

OLSR uses **multipoint relays (MPRs)** to **avoid unnecessary broadcast retransmissions**. A node periodically broadcasts a message to neighbours to **compute the MPR set** and exchange neighbourhood information. From neighbourhood information it calculates the **minimum set of one-hop relays needed to reach all two-hop neighbours** — that set is the **MPR set**.

OLSR differs from classical **link-state** protocols in two dissemination factors:

1. **Only the MPR nodes of a node A** need to forward link-state updates issued by A.
2. The **size of A's link-state update is reduced** because it consists only of neighbours that **selected A as their MPR**.

Thus OLSR **reduces** link-state protocol overhead. It is used where nodes are **densely deployed**. It calculates the **shortest path** to an arbitrary destination.

Functions of OLSR:

Neighbour sensing. Each node must certify the **nature of its link** with neighbours because of radio transmission. OLSR specifies two link types: **symmetric** and **asymmetric**. Each node sends **Hello** messages at a Hello interval to one-hop neighbours for neighbour sensing; Hellos **are not forwarded**. A Hello contains the **neighbour list and link status**, which allows reduction of the **two-hop neighbour set** and their status. **MPR selection** is then made and the list is added into Hello messages. Using this MPR list, an **MPR selector list** is constructed: neighbours that **selected this node as MPR**. Messages received from **MPR selectors will be forwarded**.

MPR flooding. The aim is to **control traffic flooding**. MPRs are selected so that a flooding message transmitted by the MPR set **reaches all two-hop neighbours**. The MPR set of node n is the **smaller subset of symmetric one-hop neighbours of n** that have **symmetric links with n 's two-hop neighbours**. MPR flooding **eliminates transmission duplication** and **minimizes reception duplication**.

Topology diffusion. The objective is to **create routing tables** using periodic **topology control (TC) messages**. TC messages are circulated by each node with a **non-empty MPR selector set** to all network nodes, advertising at least the **links between itself and nodes in its MPR selector set**. TC messages contain enough information for nodes to **construct a topology table** and then **derive a routing table**. Routes use a **shortest-path algorithm such as Dijkstra**, providing the **best possible hop count**. Routing tables must be **recalculated** whenever neighbour-link or topology information changes.

TORA — Temporally-Ordered Routing Algorithm TORA is a **hybrid** routing protocol, presented as effective against MANET limitations caused by **high node mobility**. **Congestion** is a major MANET problem. Traditional **shortest-path**, **adaptive shortest-path**, and **link-state** routing **cannot work properly** with mobile stations; it is difficult to update every dynamic node's routing tables.

Each node **broadcasts a query packet**; recipients broadcast an **update** packet. TORA supports **loop-free multiple routes** using a **flat, non-hierarchical** algorithm and provides **better scalability**. To discover a route it uses a **DAG (directed acyclic graph)** and a set of **totally ordered heights**. Information may flow in **only one direction** (unidirectional), so there is **no chance of an infinite loop**.

Three basic operations: **route creation, route maintenance, route deletion**. Three control packets: **QUERY, UPDATE, and CLEAR**.

Route creation. QUERY and UPDATE create a new route. A QUERY carries a **destination ID**. An UPDATE holds **destination ID and height** of the node. Each node maintains a **route-required flag** (initially unset) and the **time of broadcast of the last UPDATE**.

When a node has **no directed link** and an **unset** route-required flag, it **broadcasts QUERY** to neighbours and **sets** the flag. On receiving QUERY:

- If there is **no downstream link** and the flag is **unset**: **rebroadcast QUERY** and set the flag.
- If there is **no downstream link** and the flag is **already set**: **discard** the QUERY.
- If there is at least one downstream link with **null height**: **set its height** and **broadcast UPDATE**.
- If there is at least one downstream link with **non-null height**: compare the time of last UPDATE broadcast with the time the link over which QUERY arrived became active; when the link becomes active, **broadcast UPDATE** and **discard QUERY**. If the route-required flag is set, **broadcast QUERY**.

When a node receives UPDATE it updates height-array entries:

- If the route-required flag is **set**: set height, update all link-state-array entries, **unset** the flag, **broadcast UPDATE** with the new height.
- If the flag is **not set**: just update the link-state array and apply **route-maintenance** techniques.

Route maintenance. TORA maintains routes after **topological change** and **re-establishes routes within a finite time**. Maintenance is needed when a node's height is **non-null**. A neighbour with **null** height is **not considered**. Five cases:

Case	Name	Taught meaning
1	Generate new reference level	Link failure: node lost its last downstream link. If it has an upstream neighbour it updates its reference value ; otherwise it sets height to null .
2	Propagate the highest neighbour's reference level	If the node has no downstream link , it propagates the reference level to upstream neighbours .
3	Reflect back a higher sub-level	After link reversal , if downstream links fail, the node reflects back the reference height with the reflection bit set.
4	Partition detection	If the set reference value equals neighbours' reference height, the node has detected a partition, raises the route, and sets height to null .
5	Generate new reference level	New reference value when the node experienced a link failure between propagation of a reference level and reflection of a sub-level .

Route deletion. In case 4 of maintenance, a node sets height from the **direction of edges toward the destination**, updates the link-state array, and **broadcasts a control packet** (CLEAR). On receiving it:

- If the **reference level matches**, the node sets its height and **sets height for each neighbour to null**, updates link-state arrays, and **broadcasts** the control packet.
- If the reference level **does not match**, it sets height for each neighbour and updates corresponding link-state entries.

At the end, height of each node in the **partitioned** portion is set to **null** and **invalid routes are erased**.

Key terms

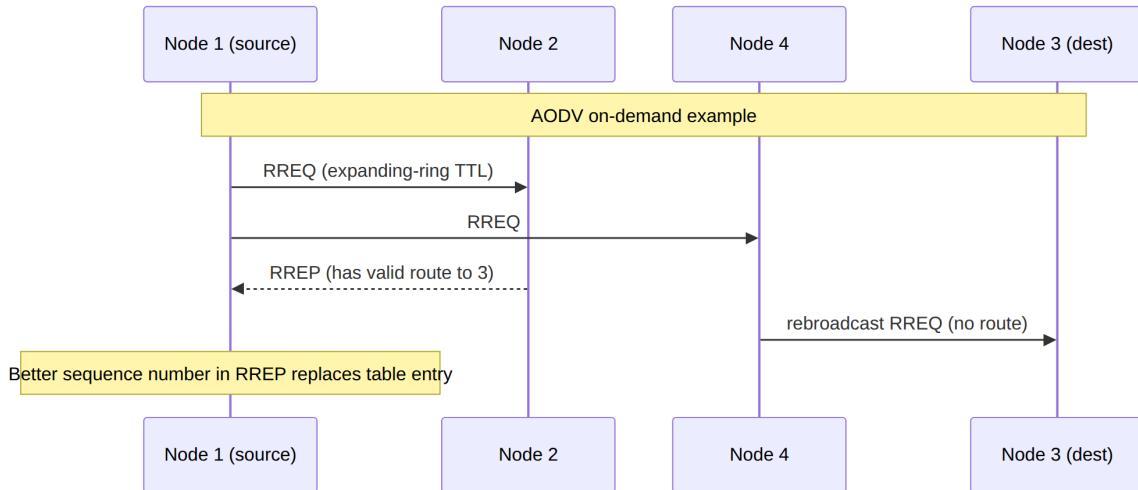


Figure 10: Architecture diagram

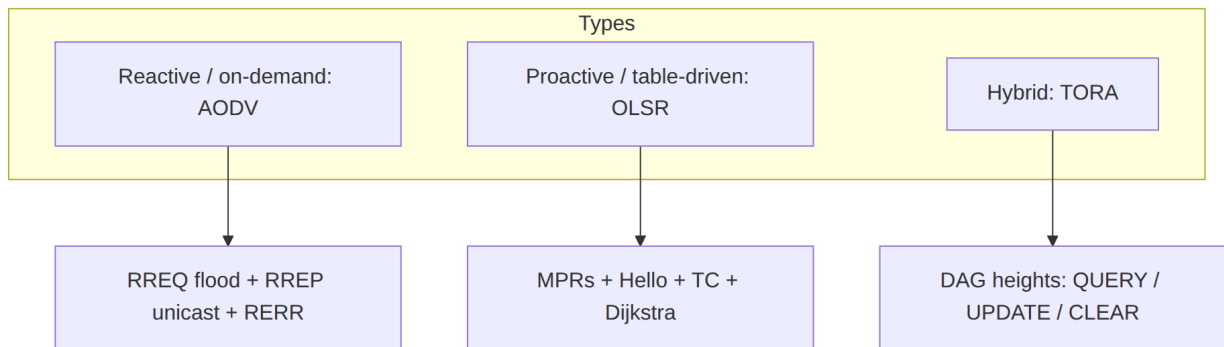


Figure 11: Architecture diagram

Term	Meaning in this lecture
Reactive / on-demand	Build a route only when needed; higher search latency
Proactive / table-driven	Continuously maintain routes to all nodes via periodic control traffic
Hybrid	Proactive backbone plus reactive flooding for extra demand
AODV	Ad hoc On-Demand Distance Vector (RFC 3561); RREQ, RREP, RERR
Expanding ring	Increase RREQ TTL on retransmission if no reply
Precursor list	Neighbours that likely use this node as next hop; used to send RERR
OLSR	Optimized Link State Routing; proactive, MPR-optimized link state
MPR	Multipoint relay: selected one-hop neighbour that forwards floods to two-hop neighbours
TC message	Topology-control advertisement from nodes with a non-empty MPR selector set
TORA	Temporally-Ordered Routing Algorithm; DAG of ordered heights, loop-free multiple paths
QUERY / UPDATE / CLEAR	TORA control packets for create, maintain, and delete routes

Lecture takeaways

- MANET routing is the algorithm-as-protocol that tracks a moving graph with chosen metrics.
- Reactive protocols save bandwidth but pay **initial delay**; proactive protocols start sessions quickly but **waste bandwidth** and recover slowly from failures; hybrid needs **foreknowledge** of size and demand.
- AODV finds a path when RREQ hits the destination or a node with a valid cache, replies along reverse routes, and uses RERR plus precursor lists when links die.
- OLSR shrinks classical link state by letting only MPRs rebroadcast and by advertising only MPR-selector links; Dijkstra then yields hop-shortest routes.
- TORA orients a DAG with heights so traffic is loop-free; QUERY/UPDATE build routes, five cases repair them, and CLEAR erases a partition.

M42: 3G Network and Security

Source: <https://www.youtube.com/watch?v=wdT6MOpPhcs> **Instructor / expert:** Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala **Course coordinator:** Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Place 3G in the ITU **IMT-2000** story (spectrum, pre-commercial and commercial launches, India's MTNL/BSNL path, and the GPRS → EDGE transition).
- State what 3G is: third-generation mobile telecommunications meeting **IMT-2000**, with at least **200 kbps** and later 3.5G / 3.75G broadband rates.
- List 3G capabilities, applications, and services (mobile Internet, HTML vs plain-text email, SMS/MMS/IM plus presence).
- Sketch 3G architecture: **RAN** (Node B, RNC) and **core** (packet domain SGSN/GGSN, circuit domain MSC, CGF) over **ATM Iu/Iub/Iur** with AAL2 voice and AAL5 data.
- Recall 3G security: CIA, KASUMI vs A5/1, IMSI/TMSI and false-base-station issues, and the **ETSI TS 121 133** threat classes.

Core concepts

History of 3G 3G is the result of ITU research and development from the **early 1980s**. Specifications and standards were developed over **15 years** and published as **IMT-2000**. Spectrum between **400 MHz and 3 GHz** was allocated for 3G.

Launches taught:

Event	Who / where	Notes
First pre-commercial 3G	NTT DoCoMo , Japan, 1998	Branded FOMA
First European pre-commercial UMTS	Manx Telecom , Isle of Man	UMTS
First commercial network "live"	SK Telecom , South Korea	CDMA2000 1x
First commercial United States 3G	Monet Mobile Networks	EV-DO
First southern-hemisphere pre-commercial demo	m.Net Corporation , Adelaide , South Australia, February 2002	CDMA2000 1x EV-DO
First 3G in Africa	MTN , 11 December 2008	UMTS on 2100 MHz
India	Government-owned MTNL in Delhi , later Mumbai	As named in the lecture
India (state operator)	BSNL , 22 February 2009 , Chennai , later nationwide	First 3G mobile provider in India After MTNL

Before 3G, a transition was made from **GPRS** to **EDGE / EGPRS (enhanced GPRS)** — described as GPRS "on steroids." **MMS boomed** and mobile Internet took off. 3G is technically a **direct upgrade to EDGE** but runs on a **wholly different set of frequencies; entirely new infrastructure** must be deployed.

What 3G is 3G is the **third generation** of wireless mobile telecommunications technology: a set of standards for mobile devices, user services, and networks that comply with **International Mobile Telecommunications-2000 (IMT-2000)** from the **ITU**.

In user terms it means **faster Internet from a mobile device**: video calls while travelling, sending images, videos, and email as fast as from a desktop. Third-generation devices and services transform wireless communications into **online, real-time connectivity** and **immediate access to location-specified, on-demand information**. Mobile phones were becoming the preferred personal-communication device, creating the world's largest consumer-electronics industry.

Applications named: wireless voice telephony, mobile Internet access, **fixed wireless** Internet access, **video calls**, and **mobile TV**.

3G is framed as an evolution of **GSM** standards. 3G networks support an information transfer rate of **at least 200 kbps**. Later releases, often denoted **3.5G** and **3.75G**, provide mobile broadband of **several Mbit/s** to smartphones and laptop modems.

Beyond 3G, the lecture predicted a landscape of technologies offering **seamless mobility** with cellular networks. **4G** would enable broadband wireless at home, office, and on the move, making web, Internet, multimedia, and entertainment services available to mobile users.

Capabilities, applications, and services **Capabilities:** high-speed data; **symmetrical and asymmetrical** data; improved voice quality; greater capacity; **multiple simultaneous services**; **global roaming**; improved security; service flexibility.

Applications listed: wireless Internet; **audio on demand**; electronic postcards; video conferencing; **secure mobile-commerce** transactions; traffic and travelling information; **location-specified 3G services**.

Services in general: wide-bandwidth services such as enhanced communication (messaging, email, video, web browsing) and **location-specific information** (stores, restaurants, gas stations, free parking nearby). Business users get **direct access to company networks** while travelling. From a marketing view, **identifying, designing, and pricing** these services is the core 3G marketing task.

Mobile Internet. The aim is to **merge cellular networks and the Internet** so handhelds (phones, PDAs) reach all Internet services. 2G was mainly **voice-centric with low data capacity**; **2.5G and 3G** raise speeds. In 3G, data speed **depends on how many users access the network at the same time**.

Email. Rated the **number-one preferred mobile service in Sweden in 2001**, followed by banking and encyclopedia use. Categories: **web-style HTML email** (more flexibility of format and appearance) vs **plain-text** letter-style messages. Email is cheap, easy, and asynchronous, but **spam / unsolicited messages** may slow mobile adoption. The lecture also cited **BBC News 2003** on the **first mobile virus**, as a reason users might avoid mobile email.

Messaging. **SMS** and **MMS** were expected to be the most utilized future mobile service, with users stepping from simple text to pictures and video. 3G makes bandwidth-

hungry video possible, but **cultural differences** matter: Europe had not adopted MMS widely (pricing, complexity); **Asians** had, on average **20-30 MMS/month** vs **1-2** for a typical European user. **Instant messaging (IM)** is popular especially among youngsters: near-real-time text, HTML, pictures, songs, video, or other files, **combined with a presence service** so users see who is available and reachable.

3G architecture A 3G wireless network consists of a **radio access network (RAN)** and a **core network**.

Core network:

- **Packet-switched domain:** **3G SGSN** and **GGSN**, providing the **same functionality as in GPRS**.
- **Circuit-switched domain:** **3G MSC** for switching voice calls.
- **Charging** for services and access through the **Charging Gateway Function (CGF)**, also part of the core.

RAN functionality is **independent** of the core. The access network gives **core-technology-independent** access for mobile terminals to different core networks and services. **Either** core domain can access any appropriate RAN service — e.g. a **speech radio access bearer from the packet-switched domain**.

RAN elements:

- **Node B** — comparable to the **base transceiver station (BTS)** in 2G.
- **Radio Network Controller (RNC)** — replaces the **base station controller (BSC)**; provides **radio resource management, handover control**, and support for connections to **both CS and PS domains**.

Interconnection inside the RAN and between RAN and core uses **Iu, Iub, and Iur** interfaces based on **ATM** as layer-2 switching. Data services run from the terminal **over IP**, which uses ATM as a reliable transport with **QoS**. **Voice is embedded into ATM** from the edge (**Node B**) and transported over ATM out of the RNC.

The **Iu** interface is split into **circuit-switched and packet-switched** parts. Iu is ATM-based: voice on virtual circuits using **AAL2**; IP over ATM for data using **AAL5**. These traffic types are switched independently to **3G SGSN** (data) or **3G MSC** (voice).

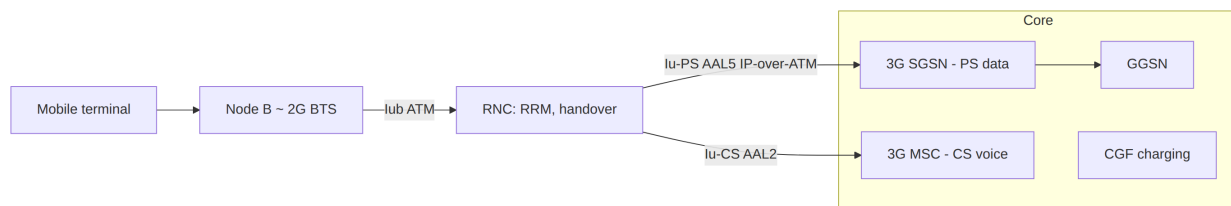


Figure 12: Architecture diagram

Effect of 4G on 3G (India-focused) In the Indian market, 4G (especially the launch of **Reliance Jio**, called the largest investment in Indian history) was the hot topic. Operators had invested significantly in **3G network and spectrum from**

2010-2015. Slow 3G uptake meant much of that investment was **not yet recovered**. Some operators consider **shutting 3G** to simplify architecture — happening in countries where 3G had run for **almost 15 years**. Developing countries such as **Indonesia and Thailand** still focused on 3G even after launching 4G, looking at **slow migration**. India was in a similar stage until 2015: **five years after launch, 3G penetration was less than 10%**. A key reason: **smartphones and 3G data pricing were both expensive**. Figures given: **300 million** Internet users, **180 million** with smartphones, but only **90 million** bought 3G plans.

Network security (general, then 3G) Network security is the **policies and practices** to prevent and monitor **unauthorized access, misuse, modification, or denial** of a computer network and network-accessible resources. Access is authorized and controlled by a **network administrator**. Users choose or are assigned an **ID and password** (or other authenticators) within their authority. It covers public and private networks used by business, government, and individuals. The most common simple protection is a **unique name and corresponding password**.

Three elements (CIA):

Element	Meaning taught
Confidentiality	Prevent unauthorized disclosure of information (including information about resources) to third parties
Integrity	Prevent unauthorized modification of resources and maintain the status quo (system resources, information, personnel); alteration may be for personal gain or revenge
Availability	Prevent unauthorized withholding of resources from those who need them when they need them

Communication infrastructure is vital and can be targeted to steal sensitive information. The lecture listed contemporary exposures: vulnerabilities exploited in **government surveillance of mobile devices, flaws in authentication protocols** that make it easy to pierce **Wi-Fi** networks, and **malware** used to control VIP communications from a PC.

3G-specific security claims. 3G offers **greater security**, allowing **mutual authentication between terminals and networks**. As 3G grows, the mobile plays a role similar to a computer: attackers can penetrate mobiles as they do PCs, and can **spy on movements, listen to calls, read messages, and access private data**. The generation is described as bringing a **vast number of vulnerabilities** — a “heaven for hackers and crackers.” Few consumers were aware of 3G threats. Each operator would need to spend **5 to 10% of its gain** securing 3G. New security solutions are needed at **service-provider and handset-manufacturer** levels. 3G makes the mobile network and bandwidth **equivalent to a computer network**, opening chances for attacks through mobile networks.

Cryptography taught: 3G uses the **KASUMI block cipher** instead of the older **A5/1 stream cipher**, but **several serious weaknesses** have also been identified in KASUMI. Increasing connectivity **grows security exposure** that is harder to manage.

Main 3G problems listed:

- **IMSI is sent in clear text** when allocating **TMSI**.
- Transmission of IMSI is **not protected**; **IMSI is not a security feature**.
- A user can be **enticed to camp on a false base station (BS)**; once on the false BS radio channels, the user is **out of reach of the paging signals of the serving network (SN)**.
- **Hijacking outgoing/incoming calls** is possible in networks with **encryption disabled**: the intruder poses as a **man-in-the-middle** and **drops the user once the call is set up**.

ETSI TS 121 133 threat classification Unauthorized access to sensitive data (violation of confidentiality):

- **Eavesdropping** — intercept messages without detection.
- **Masquerading** — fool an authorized user into believing the intruder is the legitimate system (to obtain confidential information), or fool a legitimate system into believing the intruder is an authorized user (to obtain service or confidential information).
- **Traffic analysis** — observe **time, rate, length, source, and destination** of messages to determine a user's **location** or whether an important **business transaction** is taking place.
- **Browsing** — search data storage for sensitive information.
- **Leakage** — obtain sensitive information by exploiting processes that have **legitimate access** to the data.
- **Inference** — observe a system's **reaction** to a query or signal; e.g. actively start sessions and then learn from observation of associated radio-interface messages.

Unauthorized manipulation of sensitive data (violation of integrity):

- **Manipulation of messages** — messages may be deliberately **modified, inserted, replayed, or deleted**.

Disturbing or misusing network services (denial of service or reduced availability):

- **Intervention** — prevent an authorized user from using a service by **jamming** the user's traffic, signalling, or control data.
- **Resource exhaustion** — prevent use by **overloading** the service.
- **Misuse of privileges** — a user or serving network exploits privileges to obtain unauthorized services or information.
- **Abuse of services** — abuse a special service or facility to gain advantage or **disrupt** the network.

Repudiation — a user or network **denies actions** that have taken place.

Unauthorized access to services — intruders access services by **masquerading** as users or network entities; users or network entities get unauthorized access by **misusing their access rights**.

Key terms

Term	Meaning in this lecture
IMT-2000	ITU name for 3G technical specifications
EDGE / EGPRS	Enhanced GPRS step before 3G; new 3G frequencies still need new infrastructure
RAN / Node B / RNC	Radio access network; Node B $\approx$ 2G BTS; RNC $\approx$ BSC (RRM, handover)
SGSN / GGSN / MSC / CGF	Packet core (as in GPRS), circuit MSC for voice, charging gateway
Iu / Iub / Iur	ATM RAN interfaces; Iu split CS (AAL2 voice) and PS (AAL5 data)
KASUMI	3G block cipher replacing A5/1; still taught as having serious weaknesses
IMSI / TMSI	Subscriber identity; IMSI sent in clear when assigning TMSI
False BS	Rogue base station that camps the user away from legitimate paging
ETSI TS 121 133	Specification used to classify 3G confidentiality, integrity, DoS, repudiation, and service-access threats

Lecture takeaways

- 3G is IMT-2000: ≥ 200 kbps (later several Mbit/s), new spectrum and infrastructure, not just a software bump from EDGE.
- The architecture splits RAN (Node B + RNC) from a dual CS/PS core, with ATM AAL2/AAL5 on Iu.
- User-facing 3G is mobile Internet, email (spam/virus fears), SMS/MMS/IM+presence, and location services — with sharp regional MMS uptake differences.
- Mutual authentication is an improvement over earlier generations, but IMSI exposure, false base stations, optional encryption, and KASUMI weaknesses remain.
- ETSI TS 121 133 organises 3G threats around CIA plus repudiation and unauthorized service access.

M43: 4G LTE

Source: <https://www.youtube.com/watch?v=UGQrMVqHCMQ> **Instructor / expert:** Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala
Course coordinator: Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Define **LTE (Long Term Evolution)** as the 3GPP 4G standard on the GSM → UMTS path, aimed at $\sim 10\times$ 3G speeds via new DSP and modulation, and an **IP-based** architecture with lower latency.
- Place LTE among competing 4G candidates (**UMB, IEEE 802.16 / WiMAX, 802.16m / WiMAX Advanced**) and **LTE Advanced (Release 10)**.
- Separate marketing “4G” (**HSPA**) from **4G LTE**, and state OFDM, peak-rate, and VoLTE/RCS benefits as taught.
- List LTE technical objectives (throughput, latency, simplified **E-UTRAN**, mobility) and the two resources required: a capable **network** and a capable **device**.
- Recall the lecture’s LTE security framing: confidentiality and integrity, plus research goals for comparing LTE cryptographic algorithms.

Core concepts

LTE stands for **Long Term Evolution**. It applies to **improving wireless broadband speeds** to meet increasing demand. LTE is a **4G wireless communications standard** developed by the **3rd Generation Partnership Project (3GPP)**. Engineers named it Long Term Evolution because it is the next step from **GSM (2G)** to **UMTS (Universal Mobile Telecommunications Service)** — the **3G technologies based on GSM**.

The goal of LTE was to **increase capacity and speed** of wireless data networks using new **DSP (digital signal processing)** techniques and **modulations**. It is designed to provide up to **$10\times$ the speeds of 3G** for smartphones, tablets, notebooks, and wireless hotspots.

A further goal was **redesign and simplification of the network architecture** to an **IP-based system** with **significantly reduced transfer latency** compared with 3G.

Evolution toward 4G LTE

Year / step	What the lecture placed there
1973	First call by engineer Martin Cooper from a Motorola mobile phone
1987	Cultural image: Michael Douglas in <i>Wall Street</i> on a mobile from the beach
1992	First smartphone Simon — functions beyond voice, ahead of the network’s ability to handle it
1G	AMPS — Advanced Mobile Phone System
2G	GSM and CDMA
3G	UMTS and EV-DO
4G	LTE from 3GPP, and IEEE 802.16m

IEEE 802.16m is the 4G candidate also known as **WirelessMAN-Advanced** or **WiMAX 2**.

LTE progresses through **releases**. The latest release that **qualifies as 4G** in the lecture is **Release 10**, often called **LTE Advanced**.

4G LTE is said to enable advanced technologies in **transportation, healthcare, small business, enterprise, and education**. Example: **Intuit GoPayment** — street vendors, farmer markets, and food trucks attach a **small card reader** to a smartphone or tablet so payments are quick. With large-scale **VoLTE** and **LTE Advanced** on the horizon, the instructor asked what the next forty years might hold.

LTE technical objectives Taught targets:

- **User throughput** (downlink and uplink, in MHz as labelled on the slide)
- **Downlink capacity** and **uplink capacity**
- **Latency**: user-plane transition time **less than 5 ms** in ideal conditions; **100 ms** control-plane; **fast connection setup**
- **Simplified architecture**: simpler E-UTRAN; **no RNC**; **no CS (circuit-switched) domain**; **no DCH**
- **Mobility**: optimized for **low speed** but supporting **120 km/h**

4G vs LTE vs “4G without LTE” 4G has **several competing standards**, including **Ultra Mobile Broadband (UMB)** and **WiMAX (IEEE 802.16)**. 4G technologies are designed to provide **IP-based voice, data, and multimedia streaming** at speeds of **at least megabits per second** and up to **1 Gbit/s**.

US operators named: **Verizon** and **AT&T** launching **4G LTE**; **Sprint** utilizing a **4G WiMAX** network. Most newer **Android** smartphones were 4G LTE capable; **iPhone 5** and **iPad 3** were expected to have built-in 4G LTE in the **second half of 2012**.

Speed: the difference versus “slow 3G” is **faster**. Most 4G and “true 4G” networks have **upload and download speeds that are almost identical**; **LTE is described as the fastest connection available** for wireless networks.

Two resources required for 4G LTE:

1. A **network** that can support the necessary speeds.
2. A **device** able to connect to that network and download at high enough speed.

A phone with 4G LTE inside does **not** guarantee those speeds — compared to buying a car that can do 200 mph but driving on a **55 mph** freeway.

Is LTE the same as 4G? No. There is 4G, and there is **4G LTE**. 4G is the fourth generation after 3G — faster, greater capacity. **4G LTE** is called the **purest form of 4G**. If a cellular provider describes a “4G” network **without mentioning LTE**, they are probably talking about a **High Speed Packet Access (HSPA)** network: a **faster version of 3G GSM**, still faster than 3G but **not as fast as LTE**.

Advantages of LTE; devices; generations **Advantages**: low latency; high network throughput; increased data-transfer speed; more **cost-effectiveness**; improvements over 3G.

Devices covered: mobile phones, laptops, cameras, camcorders. LTE is said to assure **interoperability with older wireless technologies: GSM, WCDMA, HSPA, CDMA, TD-SCDMA**. LTE is expected to **ensure the success of mobile Internet**.

Why 4G is faster than 3G: orthogonal frequency-division multiplexing (OFDM) — a technique for **squeezing more data onto the same amount of radio frequency**, also **reducing latency and interference**. 4G uses **different frequencies** than 3G, so the handset needs a **modem that supports those frequencies**. 4G is described as **10 times faster than then-current 3G**, so many tasks could be done through smartphones without separate broadband.

LTE peak rates taught: potential for **100 Mbit/s downstream** and **30 Mbit/s upstream**, plus reduced latency, **scalable bandwidth capacity**, and **backwards compatibility with existing GSM and UMTS**.

VoLTE (Voice over LTE): enables wireless carriers to use **IP-based 4G LTE data networks for voice calls** — WhatsApp-like call, video call, conferencing, etc.

5G (brief forward look in this lecture): fifth generation, aimed at requirements **beyond 2020**; much faster than 4G, **20 GB per second** in the wording used, fast enough to download HD movies in seconds; expected to drive **Internet of Things, smart cities, Digital India**, and a 2020 vision associated with **A.P.J. Abdul Kalam**.

Disadvantages of 4G LTE

- Advanced data rates require customers to **purchase new equipment**.
- 4G was **only presently accessible in certain cities**.
- Needs **complex hardware**.

High-definition voice, RCS, and VoLTE consumer benefits The most noticeable VoLTE benefit is **improved voice quality**. Traditional voice networks transmit calls using an **8 kbps codec**. Verizon is described as using a **13 kbps codec** with more modern compression.

Rich Communication Services (RCS), enabled by VoLTE, is a **standards-based set: video calling, file transfer, real-time language translation, video voice-mail, and instant messaging**. Many of these already exist as **over-the-top (OTT) apps (FaceTime, Skype, Google Translate)**. VoLTE's benefit is launching them **from the phone's native dialer**.

Faster call setup than older **circuit-switched 2G**.

Improved battery life versus other VoIP apps. A **Signals Research Group** report is cited: VoLTE offers much improved battery life compared with OTT VoIP such as Skype; **consistently higher call quality** than circuit-switched voice and OTT apps; and **far lower network consumption than a Skype voice call**, hence longer smartphone battery life.

Integration of VoLTE with Voice over Wi-Fi. Better integration between cellular voice and Wi-Fi calling matters as operators use Wi-Fi for some voice and data. Improving the transition between a cellular call on **licensed** spectrum and a Wi-Fi call

on **unlicensed** spectrum helps **indoor coverage** where licensed spectrum is limited. Call setup for Voice over Wi-Fi is the **same as for VoLTE**, so once VoLTE is fully deployed it is easier to offer VoWiFi where licensed spectrum is **weak or unavailable**.

True device interoperability. Once VoLTE is interoperable among carrier networks, the **CDMA/GSM divide** among the four major US operators would **no longer be an issue**. A consumer might buy **one device** that operates on any wireless network if it supports **enough LTE bands** — voice and data all on the same 4G LTE technology.

Security of 4G LTE “LTE security” is presented as a **defence mechanism versus Internet threats** from multiple attack types.

- **Confidentiality:** the **right people** can get the **right data**.
- **Integrity:** the message from party A to party B **did not change**.

These are the essential points for Next-Generation mobile security. More powerful security is to be achieved by:

- Discovering the **main factors used to compare LTE cryptographic algorithms**
- Finding **trends and weaknesses of each** LTE cryptographic algorithm
- **Evaluating which algorithm provides higher security**
- **Proposing an algorithm from the existing LTE set** that may enhance security of existing LTE cryptography

Key terms

Term	Meaning in this lecture
LTE	Long Term Evolution; 3GPP 4G on the GSM/UMTS path
LTE Advanced	3GPP Release 10 , the release taught as qualifying as 4G
E-UTRAN	Simplified LTE radio architecture: no RNC, no CS domain, no DCH
HSPA	High Speed Packet Access — “4G” branding without LTE; faster 3G GSM, slower than LTE
OFDM	Orthogonal frequency-division multiplexing; why 4G is faster
VoLTE	Voice over LTE: IP voice on the LTE data network
RCS	Rich Communication Services launched from the native dialer
WiMAX / 802.16m	Competing 4G path (WirelessMAN-Advanced / WiMAX 2)
UMB	Ultra Mobile Broadband, another competing 4G standard

Lecture takeaways

- LTE is 3GPP’s IP redesign of the GSM/UMTS family: DSP/OFDM speed, simpler E-UTRAN, sub-5 ms user-plane latency in the ideal case.
- Marketing “4G” may be HSPA; **4G LTE** is taught as the pure 4G form, still needing both network and device — and the right spectrum.

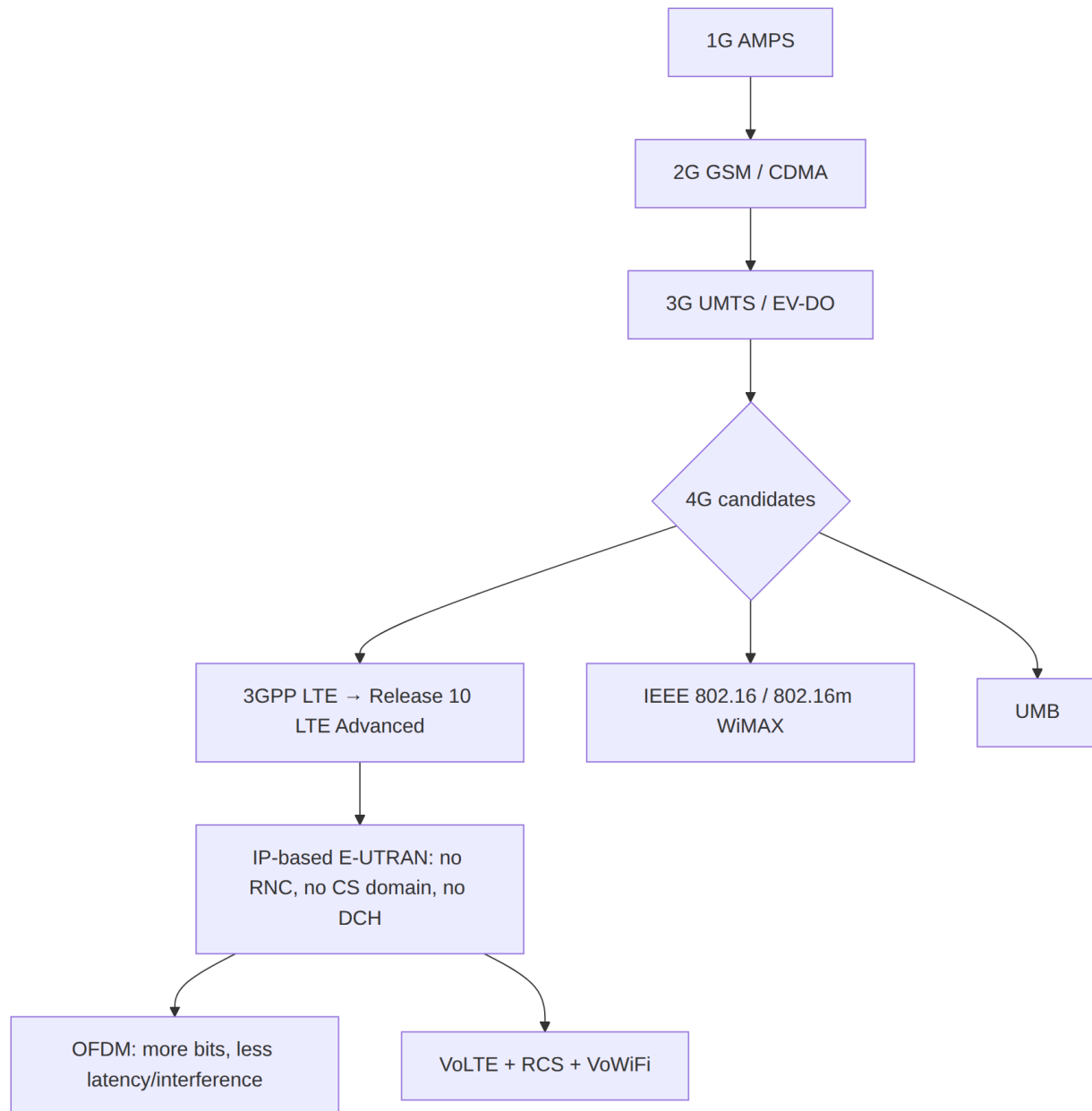


Figure 13: Architecture diagram

- VoLTE is not only HD codecs (8 vs 13 kbps) but RCS, faster setup, better battery than Skype-class OTT, and a path to VoWiFi and cross-carrier handset interoperability.
 - Security is framed as CIA plus a research programme over LTE cryptographic algorithms, not a finished protocol walkthrough in this lecture.
-

M44: 5G

Source: <https://www.youtube.com/watch?v=vtgDWzFM8Uo> **Instructor / expert:** Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala
Course coordinator: Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Define **5G** as fifth-generation mobile/wireless broadband as the lecture presented it, including the **IEEE 802.11ac** claim and the generation path from 1G through 4G LTE.
- State **NGMN Alliance** rate, connection-count, spectral-efficiency, coverage, signalling, and **~1 ms latency** requirements.
- Describe the taught **all-IP 5G architecture**: user terminal plus multiple simultaneous **radio access technologies (RATs)**, software-defined radios, and terminal-side choice via middleware.
- Contrast 5G with 4G on speed, capacity ($\sim 1000\times$), latency, millimetre-wave vs existing bands, MIMO, and small cells.
- List benefits, device/network requirements, applications (IPv6, global coverage), disadvantages, and the “nanocore / 6G” closing remarks — all as taught.

Core concepts

5G is **fifth-generation mobile networks** or **fifth-generation wireless systems**. The lecture describes it as upcoming fifth-generation mobile-network technology and as coming **wireless broadband based on the IEEE 802.11ac standard**. It is called a new mobile revolution: worldwide cellular phones, PDA-like devices, “the whole office on your fingertips.” 5G is credited with **extraordinary data capabilities**, tying together **unrestricted call volumes** and **infinite data broadcast** within the latest mobile operating systems. A dashboard-style figure compared **3G, 4G, and 5G** speeds.

Generations as told in this lecture The **G** stands for **Generation**.

- **1G** — early **1990s**, voice.
- **2G** — people could **send text messages** between cellular devices (industry-driven).
- **3G** — calls, texts, and **Internet browsing**.

- **4G** — enhanced 3G capabilities: browse, text, call, and **download/upload large video files**.
- After 4G, companies added **LTE (Long Term Evolution)**. With 4G connectivity, **LTE became the fastest and most consistent variety of 4G** compared with competing technologies like **WiMAX**. WiMAX and LTE achieved **similar outcomes**, but a **single standard** was important; **LTE makes 4G faster**.

5G vs 4G (speed/coverage). 5G will provide **better speeds and coverage** than 4G. The lecture states 5G **works with a 5 GHz signal** and is set to offer speeds of up to **1 Gbit/s for tens of connections**, or **tens of Mbit/s for tens of thousands of connections**. 5G **builds on 4G LTE**: still texts, calls, and web, but with speed that makes **Ultra HD and 3D video** download/upload easy.

NGMN Alliance requirements The **Next Generation Mobile Networks Alliance** defines requirements a 5G standard should fulfil:

- Data rates of **tens of Mbit/s for tens of thousands of users**
- **100 Mbit/s** for **metropolitan areas**
- **1 Gbit/s simultaneously** to many workers on the **same office floor**
- **Several hundreds of thousands of simultaneous connections** for a **massive wireless sensor network**
- **Spectral efficiency** significantly enhanced compared with 4G
- **Coverage** improved
- **Signalling efficiency** enhanced
- **1 millisecond latency** (or less — “virtually zero latency”); **smart communication**; latency reduced significantly compared with LTE

Design of 5G mobile network architecture The proposed model is an **all-IP** architecture for **wireless and mobile-network interoperability**.

The system consists of:

- A **user terminal** with a **crucial role**
- A number of **independent, autonomous radio access technologies (RATs)**

Within each terminal, **each RAT is seen as an IP link** to the outside Internet. There must be a **different radio interface for each RAT**. Example: access to **four different RATs** requires **four different access-specific interfaces, all active at the same time**, for the architecture to be functional.

Is 5G really faster than 4G? Yes, “much faster.” **4G LTE transfer speeds top out at about 1 Gbit/s** — about an **hour** to download a short HD movie in **perfect conditions**. People rarely see 4G’s maximum because the signal is disrupted by **buildings, microwaves, other Wi-Fi signals**, and more.

5G is said to increase download speeds up to **10 Gbit/s** — a **full HD movie in seconds** — and to **reduce latency significantly**. It is described as a path to **thousands of connected devices** in homes and offices.

Concepts for 5G mobile networks

- 5G terminals will have **software-defined radios** and **modulation schemes**, plus **new error-control schemes that can be downloaded from the Internet**.
- Development is seen as **toward the user terminal** as the focus of 5G.
- Terminals access **different wireless technologies at the same time** and should **combine different flows from different technologies**.
- **Vertical handovers should be avoided** — not feasible when there are many technologies, operators, and service providers.
- **Each network is responsible for handling user mobility**; the **terminal makes the final choice** among wireless/mobile access-network providers for a given service, based on **open intelligent middleware** in the mobile phone.
- Again: main user terminal plus independent autonomous RATs, each an **IP link** to the Internet.

How cellular radio works (baseline) and 5G spectrum A large group of global telecoms is working on **worldwide 5G standards**. Experts expect **compatibility with 3G and 4G**, but **most standards are not yet solidified**.

In the most basic form, cell phones are **two-way radios**. On a call, the phone converts voice to an **electrical signal** and transmits it to the nearest **cell site** using **radio waves**. The cell site bounces the wave through a **network of cells** to the friend's phone. The same happens for photos and video.

When a new mobile technology comes along, it is assigned a **higher radio frequency**. **4G occupied frequency bands up to 20 MHz; 5G will likely use the frequency band up to 6 GHz**. New technologies occupy higher frequencies because those bands typically **are not in use** and **move information faster**. The problem: **higher-frequency signals do not travel as far**. **MIMO (multiple-input multiple-output)** will probably be used to **boost signals** wherever 5G is offered.

Industry / investment framing 5G is an opportunity for a **more sustainable operator investment model**. Each generation unlocks value in unanticipated ways: **SMS** was expected to have negligible impact and became huge; **video calling** was expected to be the next big thing and was **slow to gain traction**. 5G is a **paradigm shift** for all mobile-ecosystem stakeholders. **Regulators** can create healthier environments that stimulate investment. Some **3G/4G business cases may not be the right ones for 5G**.

5G vs 4G (timeline and speed claims) 4G is widely deployed and still evolving; focus switches to 5G, still **several years from deployment** at lecture time, but the “buzz word.” 5G follows global adoption of **4G LTE**. Since **2012**, a new telecommunication standard is introduced **approximately every 10 years**:

Generation	Year given
2G	1991

Generation	Year given
3G	2001
4G	2012
5G	expected 2020 or 2022

5G **does not have a defined, universally agreed-upon standard**, though likely technologies are emerging. **4G connections** offer average speeds of **25-40 Mbps at home**; 5G is likely **in excess of 500 Mbps**. Many entities believe 5G will offer **1 Gbit/s mobile to 10 simultaneous connections**. Chinese vendor **Huawei** expects 5G to be at least **100 times faster** than the fastest 4G LTE then available (a 4G+ speed test vs 5G “up to 100 times faster”).

5G technologies and frequencies **4G in the UK** currently uses the LTE standard on **800 MHz, 1800 MHz, and 2.6 GHz**. The **1800 MHz** band was **repurposed by EE and Three** for LTE; other frequencies were made available to all networks in the **2012 4G spectrum auction**.

Huawei is said to believe 5G will combine **new radio access technologies (RAT)** with **existing** wireless technologies: **LTE, HSPA, GSM, and Wi-Fi**.

Samsung and researchers at **New York University** discussed **millimetre-wave** frequencies. Millimetre-wave frequencies are taught as lying between **3 and 300 GHz** (ASR “MHz” corrected: the lecture says they are **much higher** than existing network standards). Benefit: they are **scarcely used** by other broadcast technologies such as **AM/FM radio and TV**, so they have potential for **greater speeds and more data**. Downside: millimetre waves **do not pass through solid objects well** and are **difficult to sustain over long distances**. An mmWave approach would use **lots of smaller base stations**, whereas current networks combine **large base stations** and smaller antennas in dense areas. **All 5G standards** are expected to use this small-cell approach for **maximum capacity and minimal latency**.

There is **no universal 5G standard**; launch might see a **variety of standards under the 5G banner**, similar to how both **LTE** and **WiMAX** (and an “American LTE” used by **CDMA** networks) operated under **4G**.

Benefits of 5G For **businesses**: additional **capacity and speed** for greater mobile working; employees can **video-conference from any location**.

For **consumers**: e.g. **download a film to a smartphone in under a second**.

Capacity: current 4G contracts promise speeds up to **150 Mbit/s** on the go but often leave users with **inclusive data allowances of less than 5 GB** (higher allowances cost extra because of limited capacity). 5G is expected to offer **upwards of 1,000 times the capacity of 4G**, so networks could offer **greater data allowances**.

Latency: defined as time for data transfer (buffering a video, loading a page). As 4G user counts rose, so did latency (example **4G speed test: 49 ms**). **Huawei** is targeting latency of **less than 1 ms**, versus **in excess of 50 ms on 4G** — video starts

when you hit play. Reduced latency also benefits **video games, automated equipment, IoT, and augmented reality** (Oculus Rift, Gear VR, Microsoft HoloLens) by removing delays.

Coverage: depending on the standard, 5G may expand into **hard-to-reach areas**. 4G rollout in the UK still left weak spots. 5G networks could reach further by deploying **multiple smaller antennas** that emit in **multiple directions** and even **bounce off solid surfaces**.

Requirements of 5G (as listed)

- Better **coverage area** and **high data rate at the cell edge**
- **Low battery consumption** / longer battery life
- Availability of **multiple data-transfer paths**
- Around **1 Gbit/s** data rate easily possible
- **Security is more**
- Good **energy efficiency** and **spectral efficiency**
- **Worldwide coverage**
- About **90% reduction in network energy usage**

Because of these advantages, fifth-generation wireless is called **very much essential**.

Applications

- Network availability **everywhere**; use computers and mobiles **anywhere, any-time**
- Makes the world a **real Wi-Fi zone**
- **Mobile IP address** assigned depending on the **connected network and geographical position**, using **IPv6**
- **Unified global standards**
- Receive **radio signals even at higher altitude**

Disadvantages (as taught)

- Research is **still going on** / under process
- **Speed claims seem difficult to achieve** because of other devices and “incompetent technology” in other parts of the world
- **Many old devices must be replaced** (not 5G-capable), increasing cost
- **Developing infrastructure** also increases cost
- **Security and privacy issues yet to be solved**

Future span Future enhancement of “**nanocore**” will be incredible as it combines with **artificial intelligence**: control an intelligent robot from a mobile; a mobile that **automatically types what the brain thinks**. **Google Hot Trends** rated the term **6G** as the **17th most searched** word. The lecture also mentioned an “**iPod 6G**” in seven colours with an aluminium body (as spoken). Close: “that’s all for 5G technology.”

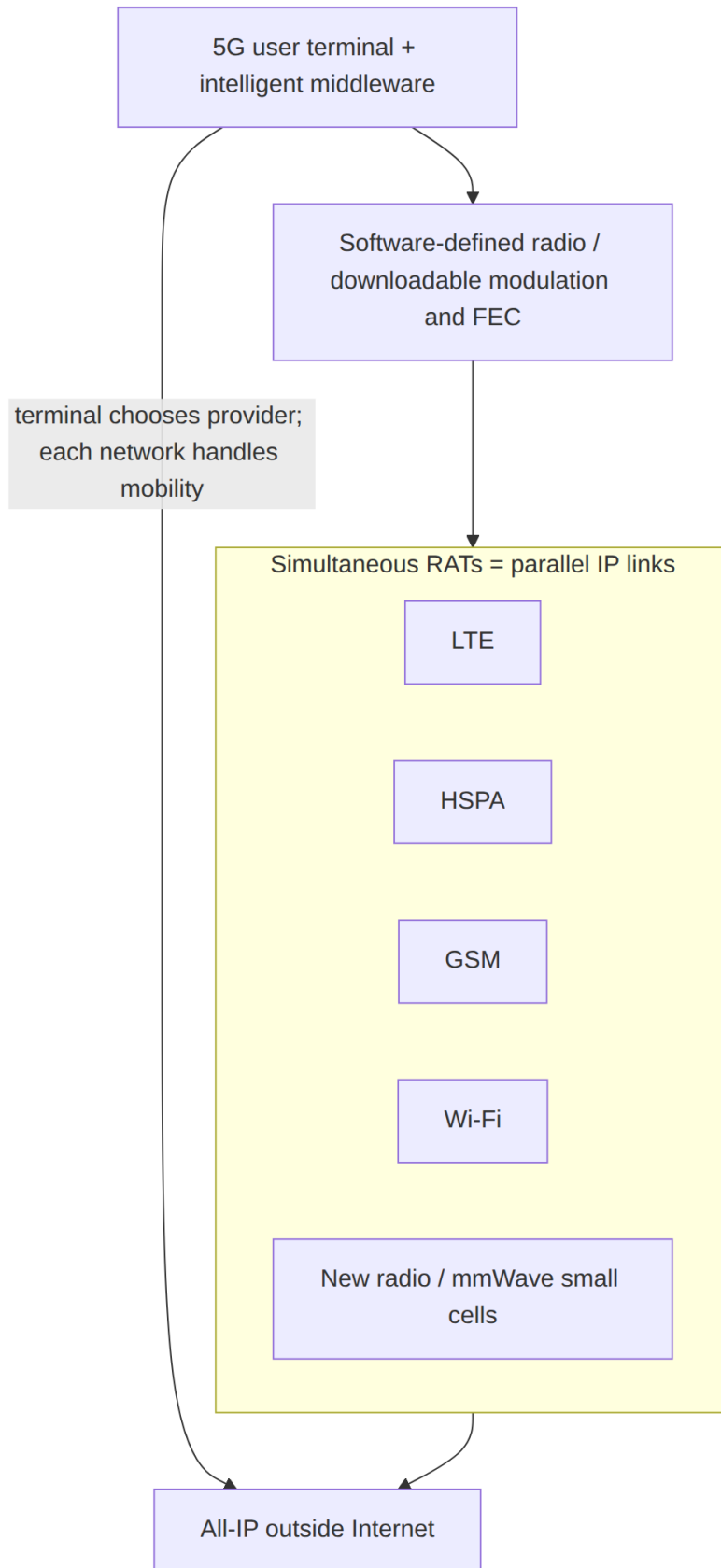


Figure 14: Architecture diagram

Key terms

Term	Meaning in this lecture
5G	Fifth-generation mobile/wireless; lecture also ties it to IEEE 802.11ac and a 5 GHz signal
NGMN Alliance	Source of 5G rate, massive-IoT connection, spectral, coverage, and ~1 ms latency requirements
RAT	Radio access technology; each appears as an IP link; multiple active interfaces at once
MIMO	Multiple-input multiple-output, to boost shorter-range higher-frequency 5G signals
Millimetre wave	Very high frequencies (taught as 3–300 GHz): fast, little broadcast coexistence, poor through walls — hence small cells
LTE vs WiMAX	Competing 4G; LTE became the consistent standard 5G builds on
IPv6	Addressing for mobile IP tied to network and geography in the 5G application vision
Nanocore	Speculative future 5G enhancement combined with AI

Lecture takeaways

- 5G is taught as all-IP, multi-RAT, terminal-centric: the phone keeps several radios up and chooses a provider; networks handle mobility; vertical handovers are to be avoided.
- NGMN targets tens of thousands of users, massive sensor connections, better spectral/signalling efficiency, and about **1 ms** latency.
- Speed/capacity claims in the lecture range from 500+ Mbps typical, 1–10 Gbit/s peaks, ~1000× 4G capacity, and Huawei's "100× fastest LTE" / sub-1 ms latency — with the caveat that **no universal standard existed yet**.
- Higher bands (up to 6 GHz and mmWave) buy rate but lose range, so MIMO and dense small cells (including bounce-off-surfaces coverage) are part of the story.
- Benefits (IoT, AR, film-in-a-second, bigger data plans) sit beside unsolved security/privacy, device replacement cost, and unfinished research.

M45: VoIP Protocols

Source: <https://www.youtube.com/watch?v=-ArpkvGMndk> **Instructor / expert:** Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala **Course coordinator:** Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Define **VoIP** as packetized digital voice/multimedia over IP instead of circuit-switched TDM, and explain why IP alone is a poor realtime transport.

- Place VoIP against the **OSI** layered model and name the three signalling/control protocols taught: **ITU-T H.323**, **IETF SIP**, and **MGCP**.
- Trace sender-to-receiver processing: codecs **G.711**, **G.729**, **G.723**, RTP/UDP/IP headers, playout buffer, and SIP/H.323 for call setup/teardown.
- List **QoS** factors: delay (source, receiver, network), jitter, packet loss, echo, and throughput.
- Recall VoIP **threat classes** (DoS, sniffing, eavesdropping, spoofing, toll fraud, SPIT) and the **defences and deployment options** the instructor listed — not attack recipes.

Core concepts

VoIP (Voice over Internet Protocol) takes **analog audio** (as on a telephone) and turns it into **digital data** transmitted over the Internet. It is the methodology and group of technologies for **voice communications and multimedia sessions over IP networks**. Voice travels in **packets** using the Internet as the medium; **IP is used rather than traditional circuit transmission**.

Voice can be digitized; digitized voice can be sent in packets. In the past, communication through a **fixed circuit-switched network** dominated; recent years emphasize **data networks**, especially VoIP. The lecture says VoIP is defined in **ITU-T H.32x** recommendations and (as spoken) **RFC 2543** (ASR “2443”; original SIP). SIP is later given as **RFC 3261**.

Why VoIP, despite IP’s weaknesses IP delivers packets carrying digitized voice, but **IP was not designed for realtime traffic** such as voice and video. IP is **connectionless**: no virtual connection is established before transmission. IP makes **no guarantees of reliability, flow control, error detection, or error correction**. Potential errors include **out-of-sequence packets** or **loss**. Voice needs a **guaranteed connection** and **reasonable delay**.

VoIP still succeeds **partly due to the high cost of traditional circuit-switched TDM**. VoIP uses a **packet-switched** network and makes the network **transparent to upper layers** involved in voice. Existing IP networks also allow **integration of voice and data**. To leverage connectionless IP, vendors developed **higher-layer protocols** to address guaranteed connection and transmission.

VoIP and the OSI model VoIP follows a **layered model comparable to OSI**. Layering provides a framework and standard, making the system **manageable and flexible**. Each layer is relatively independent; changes in one layer should have **no or minimal impact** on others.

Signalling protocols The two most generally used VoIP protocols are **ITU-T H.323** and **IETF SIP**. Both are **signalling protocols** that **discover, maintain, and terminate** a VoIP call. Additionally, **MGCP (Media Gateway Control Protocol)** provides signalling and management between **VoIP gateways** and traditional **PSTN (Public Switched Telephone Network)** gateways.

ITU-T H.323 A **comprehensive** ITU-T specification for **voice, video, and data** across a network. Sub-protocols taught:

Sub-protocol	Role in this lecture
H.225 H.235	Call control — call setup and teardown Protection framework for H.323 and call setup; security measures: authentication, integrity, privacy, some non-repudiation ; designed to work with H.245 and H.225
H.245	Media methods and parameter negotiation (terminal capabilities)
H.450	Supplementary services (e.g. call hold, telephony features)

Call setup is secured through **TLS**. Once established, call management starts so **encryption and media-channel data** can be negotiated.

H.323 uses **RTP (Real-time Transport Protocol)** or **RTCP (RTP Control Protocol)** as transport, riding on **UDP**. Encryption is performed **within the RTP packet** by third-party hardware or at the **network layer**.

Authentication under H.323 can be **symmetric encryption-based** or **subscription-based**. For symmetric encryption-based authentication, **previous contact is not required** because the protocol uses **Diffie-Hellman** to get a shared secret. Per **H.235**, subscription-based authentication **requires a previous shared secret**, in three variations:

1. **Password-based with symmetric encryption**
2. **Password-based with hashing**
3. **Certificate-based with signatures**

SIP — Session Initiation Protocol A **text-based application-layer** protocol for **signalling and session management** in the packet telephone network. Defined in **RFC 3261**. SIP uses a **request-response** model. **Authentication and authorization** are handled either by a **lower-layer scheme** or **on a request-by-request basis** with a **challenge-response** mechanism.

SIP is **lightweight**; its **own security capabilities are very limited**. Requests and responses **cannot be end-to-end encrypted** because fields such as **request and routing** must be **visible to proxy servers** present in many architectures, so requests are **routed properly**. **Voice data is transmitted in clear text over UDP** (and related IP). Although SIP supports **S/MIME-based encryption** using digital certificates, **certain header fields** in requests and responses **cannot be encrypted**. SIP therefore **depends on transport-layer security mechanisms such as TLS or IPsec** for security of the **entire message**.

MGCP — Media Gateway Control Protocol Published as **RFC 3435**. It expects MGCP messages to be carried over **secure IP connections** as in the **IPsec archi-**

ecture (RFC 2401), using either the **IPsec Authentication Header (RFC 2402)** or **IPsec Encapsulating Security Payload (RFC 2406)**. This allows **connection-less integrity, origin authentication, and optional anti-replay protection** of messages between:

- the **media gateway** that converts **circuit-switched traffic to packet-based traffic**, and
- the **media gateway controller (MGC)** that **dictates service logic**.

Processing in a VoIP system Sender side

1. Analog voice is converted to digital, **compressed**, and formatted using a voice **codec** such as **G.711, G.729, G.723**, etc.
2. Encoded voice is split into **equal-sized packets**.
3. Headers from different layers are attached: **RTP, UDP, and IP**, plus a **link-layer** header.
4. Packets are sent over the IP network to the destination.

Receiver side: depacketization and decoding. Throughout transmission, **time variation of packet delivery** (jitter) may occur. A **playout buffer** smooths playout; it introduces delay. Packets are queued for a stipulated time; packets that arrive **later than the playout deadline are discarded**.

Signalling uses **SIP** and **H.323** to set up new IP calls and to **close media streams** between clients.

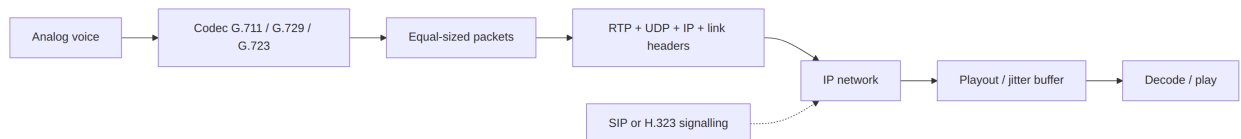


Figure 15: Architecture diagram

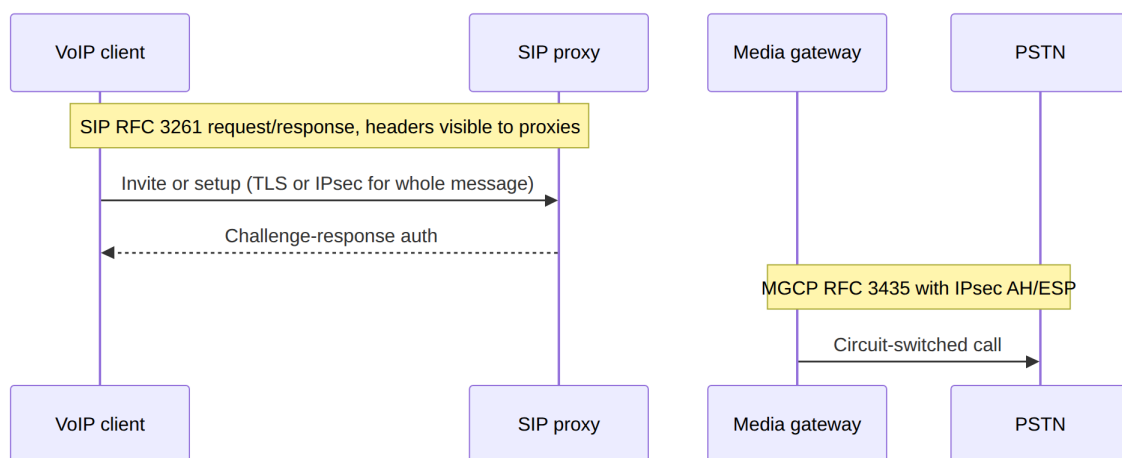


Figure 16: Architecture diagram

Quality of service (QoS) QoS measures the **degree of user satisfaction** — the network’s ability to provide services that satisfy customers. Higher satisfaction means higher QoS.

Factor	Meaning taught
Delay	Time from when one person speaks until the other hears the words. Three categories: delay at the source, at the receiver, and network delay
Jitter	Variation in transmission delay. IP does not guarantee delivery time; jitter harms voice quality
Packet loss	Packets lost, corrupted, or late. Late packets are discarded at the jitter/playout buffer or on overflow of playout or router buffers. Loss includes congestion loss plus late arrival. Retransmitting lost packets causes more delay and hurts QoS
Echo	Caller hears a reflection of their own voice . The far end may not notice. Electrical echo exists in PSTN ; acoustic echo is the difficulty in VoIP
Throughput	Maximum number of bits received out of bits sent during an interval

VoIP security — attack classes (as classified, not as recipes) VoIP, like other applications, has weaknesses that protocol designers should address before universal deployment.

Denial of service (DoS). Flooding that reduces available IP addresses, DHCP, and other **router functions**, or **blocks a server**. A VoIP-based DoS **overloads a call-processing application with so many simultaneous requests that it cannot process them**, slowing the application and **denying service to authorized users**. DoS can be directed at **any network component**.

Network sniffing. Observing network **traffic patterns**. On a **shared medium**, a user or attacker can look at others’ traffic.

Eavesdropping. Collecting sensitive information to prepare a cyber attack or gain intelligence. In VoIP, monitoring **signalling or media contents** exchanged between users, or listening in order to plan further attacks. A vendor call-manager issue was cited (Internet Security Systems): ability to **listen in or forward calls** and gain unauthorized network access.

Spoofing / caller-ID spoofing. The offender **masquerades as a licensed VoIP user** and places a call; the display looks authentic. The user may be tricked into giving sensitive information — the **VoIP version of traditional phishing**.

Toll fraud. **Unauthorized access to VoIP services**, typically for **financial gain**; among the most critical attacks for providers. Recognized via **manipulated signalling messages** or **misconfiguration of VoIP parts including charging systems**. Simple **dial-plan / open test and remote-access lines** can be abused;

after compromise, services can be **resold** with little cost to the attacker. Large-organization theft of service may go unnoticed while bills accumulate.

SPIT — Spam over Internet Telephony. Lower communication cost makes VoIP attractive to spammers versus PSTN. VoIP spam is **recorded, self-dialed** calls over IP. SPIT is **more severe than email spam** because of its **assaultive nature**, which requires a **real-time defence**. An offender may **inject a fake ID** into a call so the receiver trusts a known source and reveals **account numbers, SSN, or security-question answers**.

Security measures taught (defensive) Against DoS

- **Monitoring and filtering** — maintain a list of **suspicious users** and deny them connections/sessions
- **Authentication** — attest user identity before forwarding messages
- **Stateless proxy** — lower risk of **memory-exhaustion DoS**; can also do authentication, third-party registration, and **filtering spam sources**
- **Server design** — hardware, memory, and network affiliation as the **first line of defence**

Against eavesdropping

- Use **proper hardware**; only **authorized** access to **wiring closets** / vulnerable network points
- **Port-based MAC-address security** (e.g. on a reception/courtesy phone)
- Regularly **scan** for devices running in **unauthorized mode**
- **Encryption** of VoIP traffic

Against spoofing / masquerading

- An **effective authentication module combined with encryption**

Against toll fraud

- Providers configure **powerful firewalls** and **protect ports**

Against SPIT

- **Blacklisting** of known source IP addresses
- Vendor tools to **flag suspicious callers**
- Receivers should **not disclose important information** to unknown callers

Configuration / deployment options **USB VoIP phone adapters.** Use any **standard analog telephone** to place VoIP calls. They typically look like USB adapters with a **standard telephone port**. Once attached, the phone operates as if on the public utility.

Softphones. Software-controlled VoIP from a **PC with Internet**, using a **headset and sound card**. Providers often offer softphones **free** in exchange for using their service; users can reach dedicated VoIP phones at no extra cost.

Dedicated VoIP phones. Look like a regular telephone but connect to an **electronic network** instead of a telephone line. May be a phone plus a **base station** on the Internet, possibly on a **local wireless network**. Need a **provider and service setup**.

Dedicated routers. Connect **standard phones** to the Internet (router on **ADSL/cable modem**) and still act as an **IP router** for a PC. Providers configure them for a charge; they can work **independent of any PC or software**.

Wireless compatibility. Mobiles and smartphones can use VoIP if a wireless LAN is installed; **applicable security such as a firewall or encryption** is needed.

Issues of wireless VoIP (mainly corporate LANs, not homes): **scalability** problems for enterprises; **QoS poorer** than wired networks; **higher setup and maintenance cost** because of many access points in a limited area; **higher security threat**.

Limitations of VoIP Popularity depends on key issues — some because **IP was designed for data packets**, some because vendors do not meet standards:

Quality of service. IP design does not guarantee **realtime** voice. IP was designed for data that gets **error-free, ordered delivery**. Acceptability depends on delay **not exceeding a threshold**. **Prioritizing voice packets** can guarantee good voice quality.

Interoperability. PSTN signalling must be **interchanged** with VoIP signalling for VoIP to be common. Acceptable mechanisms named: **H.323, SIP, and MGCP**, each integrating **data, voice, and video over the same wire**.

Security. The backbone is the **Internet**, not a very secure medium: **interception of calls, DoS, fraud**. Security can use **tunnelling protocols such as Layer 2 Tunneling Protocol (L2TP)** and **encryption in SSL**, but **encryption is not widely available for VoIP**.

Integration with PSTN. VoIP works **in union with PSTN** so they appear as **one network**. A VoIP telephone number has an **IP address**; when a VoIP phone engages in a call, the **IP address is translated into the telephone number** and handed to the PSTN. Both are needed because **not everyone has switched to VoIP**.

Scalability. Lower cost and improving quality drive a high rate of VoIP users; the main obstacle is **scalability**.

The instructor noted that VoIP protocols would be discussed **in more detail in forthcoming lectures**.

Key terms

Term	Meaning in this lecture
VoIP	Packet voice/multimedia over IP instead of circuit TDM
H.323	ITU-T umbrella (H.225, H.235, H.245, H.450) with RTP/RTCP over UDP and TLS setup
SIP	IETF RFC 3261 text request/response signalling; needs TLS/IPsec because proxies must see headers

Term	Meaning in this lecture
MGCP	RFC 3435 gateway control; IPsec AH/ESP between media gateway and MGC
G.711 / G.729 / G.723	Example voice codecs on the send path
Playout buffer	Receiver jitter buffer; late packets discarded
SPIT	Spam over Internet Telephony — real-time voice spam
Toll fraud	Unauthorized use of paid voice service, often via signalling or configuration abuse
PSTN	Public Switched Telephone Network; still required for number translation and non-VoIP users

Lecture takeaways

- VoIP wins on TDM cost and voice-data integration, but IP is connectionless and offers no realtime guarantees — hence codecs, RTP, jitter buffers, and signalling (H.323, SIP, MGCP).
- H.323 is a full ITU stack with H.235 authentication variants and Diffie-Hellman; SIP is lightweight and **cannot hide routing headers from proxies**, so TLS/IPsec carry the security burden; MGCP assumes IPsec to the PSTN gateway.
- QoS is delay, jitter, loss, echo (electrical vs acoustic), and throughput — not a single number.
- Threats taught are DoS, sniffing, eavesdropping, caller-ID spoofing/phishing, toll fraud, and SPIT; responses are authentication, encryption, filtering/blacklists, firewalls, physical/port security, and user caution — not protocol exploits.
- Deployment spans USB adapters, softphones, dedicated phones/routers, and wireless LAN, with scalability, PSTN interworking, and incomplete encryption as the practical limits.

M46: Introduction to DDoS

Source: <https://www.youtube.com/watch?v=jEYv9HSp8lU> **Instructor / expert:** Dr Abhinav Bhandari, Department of Computer Engineering, Punjabi University, Patiala
Course coordinator: Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Distinguish **DoS** from **DDoS** and place both as attacks on **availability** in the CIA triad.
- Explain why Internet growth and a security-light original design made flooding and related availability attacks severe.

- State attacker **motives** and organisational **consequences** as listed in the lecture.
- Name the **protocol/architecture weaknesses** the instructor blamed (stateless forwarding, lack of authentication / IP spoofing, predictable TCP handshake, mimicry of legitimate users).
- Classify DDoS **as taught**: physical (layer 1), volumetric (layers 2-4), and application-layer (layers 5-7) — class names only, not operational recipes.

Core concepts

Internet usage grew tremendously in the last decade; lives became dependent on it for information, business, commerce, communication, education, entertainment, and research. The Internet's original goal was **unobstructed flow of data and scalability, without keeping security in mind** — now a precarious situation. A simple architectural design eased usage but **fascinated attackers** who exploit vulnerabilities for many reasons.

Those threats can target the **CIA model: confidentiality, integrity, and availability**. **Availability** is primarily targeted by **denial-of-service (DoS)** attacks: an **explicit attempt to prevent legitimate users from using a service**.

DoS has overtaken other concerns as arguably the **most severe form of attack in the last decade**. The major goal is to **disrupt or deny services**. Example: a reputed online shopping site suddenly unavailable because a **flood of packets** is surging in — one of hundreds or thousands of victims of a **pervasive, growing** Internet threat.

Traditionally the security community focused on **unauthorized disclosure or modification of information** and perhaps **theft of services**. DoS was largely **ignored as unlikely** because “the characters would not gain anything.” That is **not the case today**. DoS can put merchants out of business, cause **major visible disruption**, be used against companies from **grudge or paid attack**, or be used by **terrorists** against critical infrastructure.

A DoS aims to **deny legitimate users access to shared services or resources**. It may include attempts to **flood a network, disrupt connections between two machines, or disrupt service to a specific system or person**, preventing legitimate traffic from reaching a server, service, or website. Such an attack can be configured in many ways, targeting **operating systems** or **network services**.

The lecture contrasted **two high-level forms** (classification, not a playbook):

1. **Vulnerability / crafted-packet DoS** — crash a system by exploiting **software vulnerabilities** on the target (example class named: **Ping of Death**, using oversized **ICMP** ping traffic that some operating systems mishandled via **buffer overflow**, leading to crash, freeze, or reboot). The instructor said this form can be **prevented by patching**.
2. **Flooding DoS** — **massive volumes of useless traffic** occupy resources that would have served legitimate traffic. This form **cannot be easily prevented**; targets can be attacked **simply because they are connected to the public Internet**.

DoS versus DDoS When attack traffic comes from **multiple sources**, it is a **distributed denial-of-service (DDoS)** attack. Multiple sources **amplify** power and make **defence much more complicated**.

Both are huge threats; DDoS is **more complex and harder to solve** because:

- It can use a **very large number of machines** — a powerful weapon; **any target, however well provisioned, can be taken offline**. Gathering a large “army” of machines has become simple because **automated tools** exist that **do not require sophistication**. Even if attacking machines could be identified, **action against a network of (e.g.) 1,000 hosts** is hard.
- Some DDoS uses **seemingly legitimate traffic**: resources are consumed by a large number of **legitimate-looking messages** with **no distinguishing feature**. Because the attack **misuses a legitimate activity**, response **without also disturbing legitimate activity** is extremely hard.

Botnet roles (architecture as taught) Operating systems and network protocols were developed **without security engineering**, leaving many **insecure, unpatched machines**. Attackers implant programs on those machines. Depending on sophistication, compromised machines are called **masters, handlers, or zombies**, collectively **bots**; the attack network is a **botnet**.

Control instructions go to **masters**, which communicate them to **zombies**. Zombie machines then send attack packets that **converge at the victim or its network** to exhaust **communication or computational** resources.

Incidents are common against clients, businesses, ISPs, and well-known companies such as **Yahoo, Twitter, Facebook, eBay, Google**. A table of incidents from **2000 to 2014** was shown. Security reports after **2014** showed further strengthening versus pre-2005. **Attack size** taught: **less than 10 Gbps** before 2005; about **100 Gbps** in 2010; up to **400 Gbps** in 2014.

Motives

Motive	As taught
Recognition	Early DDoS as proofs of concept or pranks; taking a popular site offline brings underground recognition
Ideological activism	Political disagreement with an organisation, media site, corporation, or government
Financial gain	Monetary motive; after an attack they may demand ransom to prevent further attacks; these attackers are described as most experienced
Vandalism	Destroy available resources; “don’t believe in anything”; reports of increase
Revenge / competitive spirit	Business revenge; possibly dishonoured employees

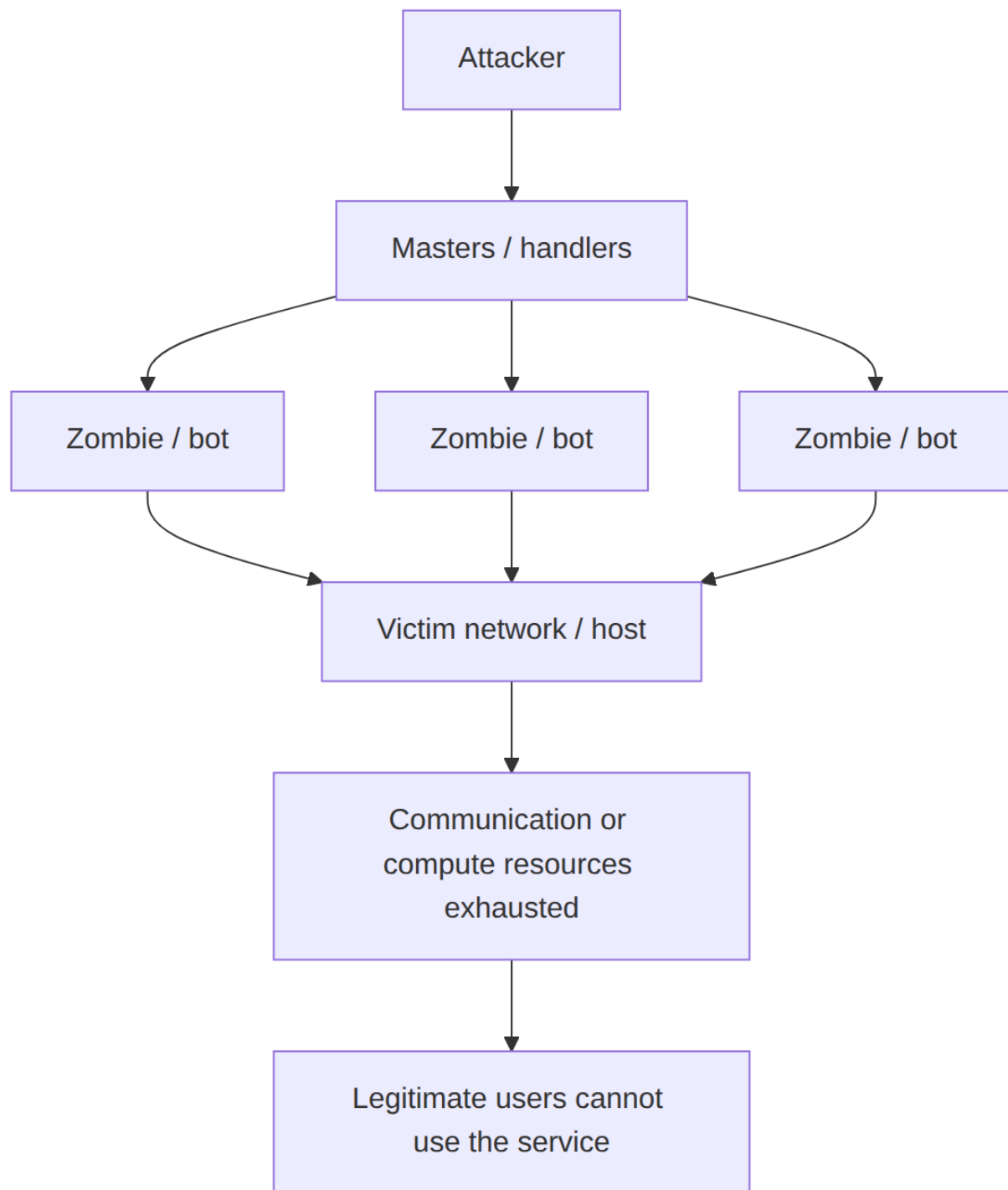


Figure 17: Architecture diagram

There is a **major lack of data** on perpetrators and motives; the **vast majority of attacks are not reported**.

Consequences Severity depends on **kind of attack, organisation, and target**. **Long-duration** attacks are more rigorous and must be **mitigated quickly after detection**. Listed consequences:

- Huge **revenue losses**
- **Reputation** damage
- Loss of **shareholder confidence**
- **Customer dissatisfaction**
- Loss of important data due to **system crashes**
- Extra **operational cost**

These can imply **social, financial, and legal** effects.

Vulnerabilities in basic protocols and architecture

1. **Stateless nature of the Internet** — intermediate routers **do not maintain state** about forwarded packets, so **accountability / forensics** are difficult and theft can go unidentified.
2. **Lack of authenticity** — without authentication, users can **claim others' identities**. The common example class is **IP spoofing** (forged source addresses).
3. **Deterministic / predictable Internet protocols** — not always a “design flaw”; often needed for proper operation. Example: **TCP connection establishment and control**. The **three-way handshake** is **predictable** and can be abused for flooding classed as **SYN flooding**.
4. **Mimicking legitimate user behaviour** — leads to **application-layer DoS**, which is hard to separate from real users.

Classification of DDoS (by OSI region) DDoS may occur at **every OSI layer**. The lecture's figure grouped three classes: **physical-layer, volumetric, and application-layer**.

Physical-layer (layer 1) Concerned with **physical security**: destruction, obstruction, malfunction, or manipulation of **physical assets**. Communication relies on nodes, hosts, switches, hubs, routers, servers, and wired or wireless media. If devices that grant service go down, service is disrupted. Attackers may deny service by **compromising physical security** so signals never arrive.

Named subclasses:

- **Destruction of assets** — damage physical assets or cut **LAN backbone** wiring so assets become unresponsive. Prevention taught: **physical security of assets**.
- **Jamming** — primarily in **wireless sensor networks**: noise close to the network so **throughput falls**; packets corrupted by **interference**. (Class of radio disruption, not a construction guide.)

- **Tampering** — interfere with node function by erasing programs or modifying code; **cryptographic keys and other sensitive information** may be extracted from captured nodes. Prevention taught: keep packages **tamper-proof**.

Volumetric (layers 2-4) Congestion by consuming **bandwidth** through flooding classes such as **MAC flooding, ICMP flooding, UDP flooding, SYN flooding**. High traffic; many DDoS variants sit here. The protocol idea of the class is to **congest the network or target with large amounts of packets** so resources are consumed, the machine may become **unresponsive** or **crash**. Traffic is described as coming through **distributed bots** remotely controlled by **bot masters**. A **key feature**: most IP packets used have **spoofed source addresses**, which **hinders traceback**.

Named volumetric classes (what they abuse, not how to run them):

Class	Lecture's characterisation
MAC flooding	Forged / cloned MAC addresses and large frame volume toward switches ; non-existent MACs used to disturb ARP caches . Switch memory is exhausted allocating resources for forged MACs, denying legitimate requests . May also disturb router tables so routing becomes unavailable
ICMP flooding	ICMP is the error-control / diagnostic protocol at the network layer. Query traffic encapsulated in IP is sent in volume with spoofed sources ; the server's replies consume bandwidth . Fragmented ICMP at high rate can overload reassembly. Named related examples: Smurf and Ping of Death (ICMP-based classes)
TCP / SYN flooding	Abuses the TCP three-way handshake . Incomplete connection requests leave the server holding half-open state until timeout , so a large number of such requests make the target unavailable. Related names: SYN flooding, fragmented acknowledgement packets as TCP-flood examples
UDP flooding	UDP is connectionless . Volume from distributed bots exhausts bandwidth. Echo and chargen services are named as commonly abused. Other UDP-related classes named: DNS flood, UDP fragmentation, VoIP flood

Application-layer (layers 5-7) In contrast to volumetric attacks, these generate **low traffic at the network** but **large request load on the web server**, which can **overwhelm server resources**. Named examples: **HTTP GET and POST request flooding**. Application protocols exploited: **HTTP, HTTPS, DNS**, and others.

Closing DoS/DDoS is a **devastating** class of cyber attack. Many **commercial solutions** exist, but an **ideal solution is still elusive**. The next lecture covers **approaches to tackle the DDoS problem**.

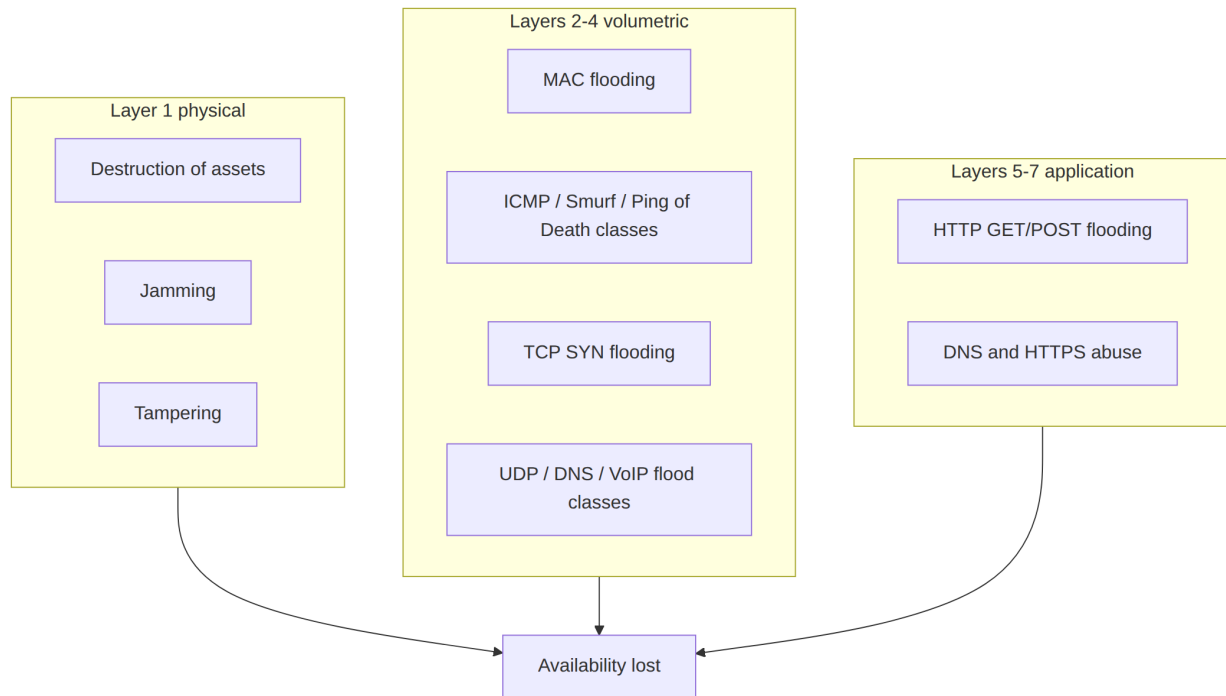


Figure 18: Architecture diagram

Key terms

Term	Meaning in this lecture
DoS	Denial of service: prevent legitimate users from using a service (availability)
DDoS	DoS whose traffic comes from many sources , amplifying impact and complicating defence
Bot / botnet	Compromised machines (masters, handlers, zombies) used as a distributed attack network
IP spoofing	Forged source addresses; hides origin and defeats simple identity-by-IP defences
SYN flooding	Volumetric class that leaves TCP connections half-open until timeout
Volumetric attack	Layers 2-4 congestion/bandwidth exhaustion
Application-layer DDoS	Layers 5-7: low network volume, high application/request load
Ping of Death	Named vulnerability class: oversized ICMP mishandled by some OSes; patchable
CIA	Confidentiality, integrity, availability — DDoS targets availability

Lecture takeaways

- Availability was historically under-weighted; flooding proved it can bankrupt, embarrass, or politically coerce a victim.
 - DDoS is harder than DoS because of **scale**, **legitimate-looking traffic**, and **spoofed identity**.
 - The Internet's **stateless forwarding**, **weak authentication**, **predictable TCP handshake**, and **easy mimicry of users** are the taught root causes — not a single bug.
 - Classification is **physical vs volumetric vs application-layer**; named floods (MAC, ICMP, SYN, UDP, HTTP GET/POST) are **types**, not procedures.
 - Patching stops some vulnerability DoS; **public-Internet flooding has no easy preventative**, and commercial tools have not closed the problem.
-

M47: Defence to DDoS Attacks

Source: https://www.youtube.com/watch?v=w_XZu6Rc0jM **Instructor / expert:** Dr Abhinav Bhandari, Department of Computer Engineering, Punjabi University, Patiala
Course coordinator: Dr Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala

Learning objectives

- Explain why **DDoS flooding** is hard to defend: easy launch, traffic similarity, IP spoofing, huge volume, many zombies, and Internet **hub-and-spoke** choke-points.
- Separate **technical** vs **social** defence challenges (distributed response, missing incident data, no benchmarks, large-scale testing; deployment economics).
- List **goals of an ideal defence**: effectiveness, completeness, minimum collateral damage, low false positives, low cost.
- Name the four **defence modules**: prevention, detection, source identification, and reaction — as strategies, not attack methods.
- Recall the instructor's **hardening tips** (filters, patches, unused services, quotas, baselines, physical security, integrity tools, redundancy, backups, passwords).

Core concepts

The previous lecture covered DDoS basics, motives, and classification. This lecture covers **why defending is difficult**, **challenges and goals for an ideal defence**, and **tips to reduce exposure**.

Why the DDoS problem is hard to solve Flooding-type DoS/DDoS is **very effective for attackers** and **extremely challenging for defence** because of these characteristics:

Simplified procedures for launching. Many DDoS tools can be obtained and set in motion. They make **agent recruitment and activation automatic** and can be used by **inexperienced users**. Some tools have existed for years and still generate effective attacks with little tweaking. (The lecture names the existence of tools as a defence challenge — it does not teach their use.)

Traffic variation / similarity to legitimate traffic. Unlike threats that need specially crafted packets (the lecture's examples of other threats: **intrusions, worms, viruses**), flooding needs **high volume** and can **vary packet contents and header values**. Separation and filtering are therefore **extremely hard**.

IP spoofing. Attack traffic can **appear to come from numerous legitimate clients**. That **defeats resource-sharing approaches that identify a client by IP address**. If spoofing were eliminated, aggressive senders might be distinguished and filtered. With spoofing, the victim sees many **service-initiation requests from seemingly legitimate users**. Ongoing sessions with known users might be told apart, but the victim **cannot distinguish new legitimate requests from attack ones**, so **no new users can be served** during the attack. Long attacks obviously damage the victim's business.

Huge volume of traffic. High volume **overwhelms the targeted resource** and makes **traffic profiling hard**. At high packet rates a defence can do only **simple per-packet processing**. The main challenge is to **discern legitimate from attack traffic at high packet speeds**.

Many zombie machines. Strength lies in **numerous agent machines distributed over the Internet**. The attacker can overwhelm even large networks, **vary** the attack by using **subsets of agents** or **few packets from each**, and thereby **defeat many traceback defences**. Even without variation, **knowing 10,000 attacking identities barely helps stop the attack**. The situation would be simpler if so many agents could not be recruited. Growth of Internet hosts and **novice users** suggests the **pool of potential agents will only increase**. **Distributed Internet management** makes wide deployment of any security mechanism unlikely; even a permanent host-hardening method would take **many years** to dent the DDoS threat.

Weak spots in Internet topology. Today's **hub-and-spoke** Internet has a handful of **highly connected, well-provisioned hubs** that relay traffic for the rest. They are built for heavy traffic, but if those few spots were **taken down or heavily congested**, the Internet would **grind to a halt**. Heavy traffic through those hotspots would have a **devastating global effect**.

In a nutshell: flooding DDoS is framed as a "**perfect crime**" on the Internet: tools and agents are abundant; sufficient volume brings the strongest victim down; the right mixture plus spoofing **defeats filtering**; businesses that rely on online access suffer considerable damage; spoofing, numerous agents, and **lack of automated tracing** grant **little risk of the perpetrator being caught**.

Challenges of DDoS defence Challenges fall into **technical** and **social** categories. Technical issues involve **current Internet protocols** and **threat characteristics**. Social issues concern how a technical solution is **introduced, accepted, and widely**

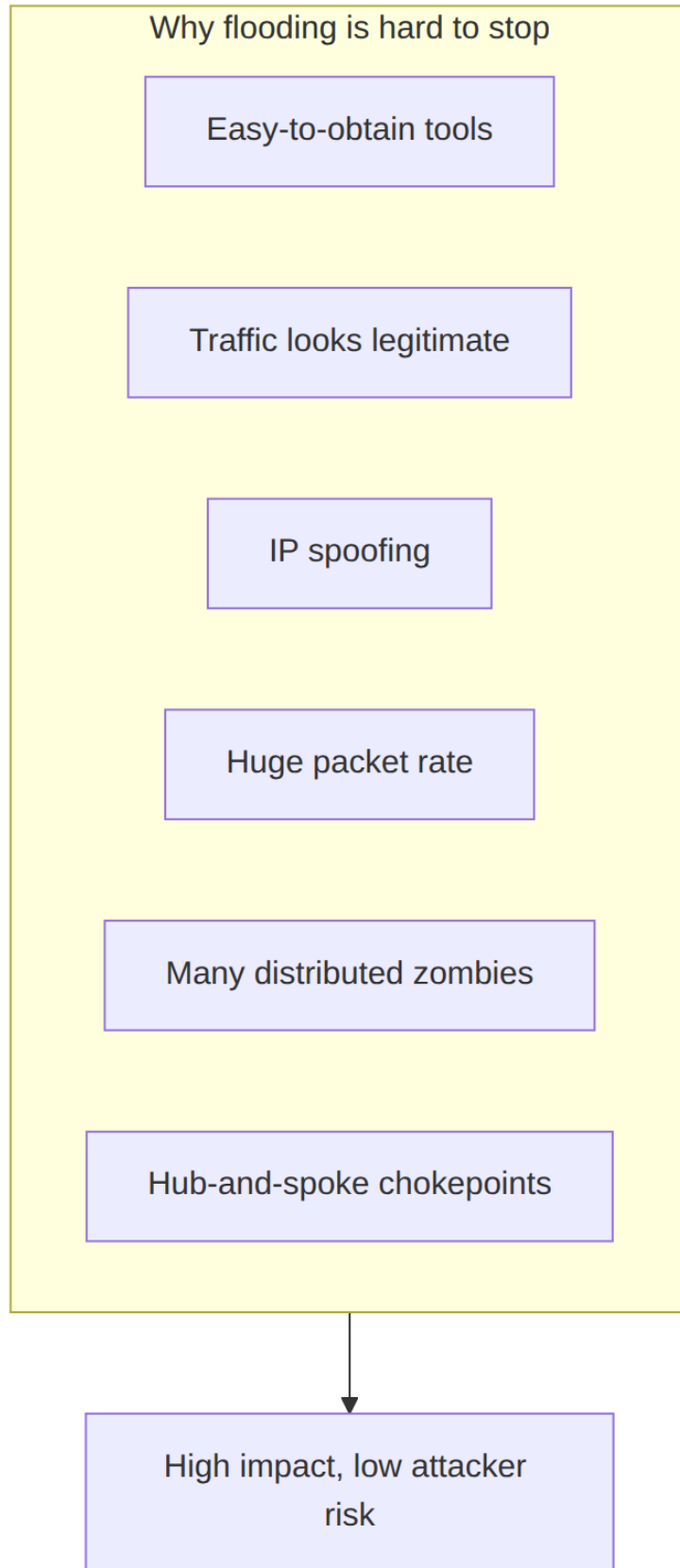


Figure 19: Architecture diagram

deployed.

The main problem in **both** is that large-scale DDoS is a **distributed threat** needing a **merit of overlapping solutions** spread across the Internet, because attacking machines may be anywhere. Attack streams can be controlled only if there is a **point of defence between agents and victims**.

Two deployment sketches:

- **One defence close to the victim** monitoring all incoming traffic — deficient mainly because it must **efficiently handle huge volume**.
- **Distributed defence** dividing workload — must be **widespread** to cover any agent-victim pair. Because widespread deployment **cannot be granted**, the technical challenge is defences that still work if **sparsely deployed**. The social challenge is an **economic model that motivates large-scale deployment**.

Technical challenges (summarised list) **Need for distributed response at many Internet points.** Very few DDoS types can be handled **only by the victim**. Response should be **distributed, possibly coordinated**, and deployed at **many points** to cover diverse agents and victims. The Internet is **administered in a distributed manner**, so wide (or even cooperative) deployment **cannot be enforced**. That **discourages researchers** from considering distributed solutions.

Lack of detailed attack information. Reporting attacks is believed to **damage the victim's business reputation**, so limited information exists; incidents often go only to **government organisations under secrecy**. It is hard to design imaginative solutions without familiarity. **Attack information must not be confused with attack-tool information** (publicly available). Useful incident data would include **type, time and duration, number of agents (if known), attempted response and its effectiveness, and damage suffered**.

Lack of defence-system benchmarks. Vendors make bold claims of completely handling DDoS. There is **no standardized testing approach** for comparison. Two harmful effects: designers present only **advantageous tests**, and researchers **cannot compare actual performance** — only comment on design.

Difficulty of large-scale testing. Defences need a **realistic environment**. That is currently impossible due to lack of **large-scale testbeds, safe live distributed Internet experiments, or simulation tools supporting several thousand nodes**. Performance claims based on **small-scale tests** are **not credible**. Some testbeds in use: **PlanetLab** and **Emulab** (ASR “IMU lab”).

Social challenges (deployment patterns) Many defences need specific patterns to work:

- **Complete deployment** — at each host, router, or network
- **Continuous / neighbour deployment** — at hosts or routers that are **directly connected**
- **Large-scale widespread deployment** — majority of Internet hosts

- **Complete deployment at specified points** — a carefully selected set; **all** must deploy to achieve the desired security
- **Modification of widely deployed Internet protocols** such as **TCP/IP** or **HTTP**

Goals of a strong DDoS defence Effectiveness. A good defence should **actually defend**: either **effective prevention** that makes attacks impossible, or **effective reaction** so the DoS effect **goes away**. For reactive mechanisms, response should be **sufficiently quick and automated** that the target does not suffer seriously.

Completeness. Handle **all possible attacks**, or at least a **large number**. A mechanism that handles **TCP SYN flooding** but cannot help on a **ping flood** is less valuable than one that handles **both**. Example: **TCP SYN cookies** help but are **not sufficient unless coupled with other mechanisms**. Completeness is also required in **detection and reaction**: if detection misses a pattern, **no response is invoked** and the attack succeeds. Completeness is **extremely hard** because attackers develop **new types designed to bypass existing defence**. **Flash events** need **clear-cut distinction** from DoS.

Minimum collateral damage. The core goal is **not merely to stop attack packets** but to ensure **legitimate users can continue normal activity despite the attack**. Legitimate traffic may share **sources** with attack traffic, share the **victim's network as destination**, share **some Internet path** with attack traffic, or share **application protocol or destination port**. **None of these legitimate categories should be disrupted**. In practice, defences cannot characterise attack traffic perfectly and **drop some legitimate traffic** — that dropping is **collateral damage**. Because attackers **conceal traffic in the legitimate stream**, resemblance is common; the problem is **real and serious**, especially with **flash events**. Protecting legitimate traffic is very important.

Low false-positive rates. Preventive mechanisms should not **hurt other forms of traffic**. Reactive mechanisms should activate **only when an attack is actually underway**. Detecting an attack when **none is happening** is a **false positive**; it causes collateral damage by triggering filtering and incurs **CPU, memory, and delay** cost on all packets.

Low deployment and operational cost. Cost must be **commensurate with benefits**. Commercial solutions have **hardware/software purchase** cost and **system-administration** cost. Some mechanisms need **ongoing administration** (e.g. **signature** systems needing updates as new attacks are characterised). Other overheads: **stateful inspection** may delay each packet; **throttling** suspicious sources may slow **legitimate** interactions with those sources. Unless costs are **extremely low or rare**, they must be balanced against the degree of protection.

Four defence modules **1. Attack prevention** — stop attacks **before they reach the target**. For DDoS that uses **spoofed traffic**, a highlighted approach is **filtering spoofed packets close to or at the attack sources** — taught as one of the **most effective** approaches for that subclass.

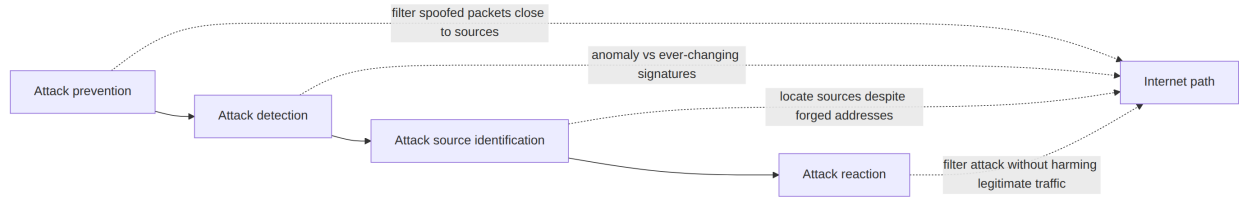


Figure 20: Architecture diagram

2. Attack detection — detect DDoS **when it occurs**; an important procedure to **direct further action**, based on **traffic-feature distribution**. **Signature-based detection is not applied** for DDoS defence because signatures **change constantly**. **Anomaly-based detection** is suitable.

3. Attack source identification — locate sources **regardless of whether the packet source-address field contains erroneous information**. Crucial to **minimize damage** and **deter potential attackers**.

4. Attack reaction — **eliminate or curtail** attack effects. The **final step**, determining overall defence performance. The challenge is to **filter attack traffic without disturbing legitimate traffic**.

Practical tips to reduce exposure Taught as administrative/hardening guidance:

1. **Implement router filters** — lessen exposure to certain DoS attacks and help prevent users **on your network** from effectively launching certain DoS attacks.
2. **Install patches** (when available) to guard against **TCP SYN flooding** — substantially reduces exposure but **may not eliminate risk entirely**.
3. **Disable unused or unneeded network services** — limits an intruder's ability to use those services for DoS.
4. **Enable quota systems** if the OS supports them (e.g. **disk quotas** for all accounts, especially those running network services). If the OS supports **partitions/volumes**, **partition the file system** to separate critical functions from other activity.
5. **Observe system performance and establish baselines** for ordinary activity; use baselines to notice unusual **disk, CPU, or network** levels.
6. **Routinely examine physical security** relative to current needs: servers, routers, unattended terminals, network access points, wiring closets, environmental systems (air and power), and other components.
7. Use **Tripwire or a similar tool** to detect changes in **configuration or other files**.
8. Invest in **redundant and fault-tolerant** network configurations.
9. Establish and maintain **regular backup** schedules and policies, particularly for **important configuration information**.
10. Establish and maintain appropriate **password policies**, especially for highly privileged accounts (**Unix root** or **Microsoft Windows administrator**).

Closing DDoS remains a **devastating** attack class; **defence is still elusive** despite many **academic and commercial** solutions. This lecture covered **why the problem is hard, challenges and goals** for an ideal defence, and **system-management tips**.

Key terms

Term	Meaning in this lecture
IP spoofing (defence view)	Makes new legitimate clients indistinguishable from attack initiators
Collateral damage	Legitimate traffic dropped or delayed by a defence
False positive	Defence treats a non-attack (e.g. flash crowd) as DDoS
Flash event	Sudden legitimate load that must be distinguished from DoS
SYN cookies	Example partial preventive for SYN floods; not complete by itself
Anomaly-based detection	Preferred DDoS detector; signatures change too often
Source identification	Finding origins even when addresses are forged
PlanetLab / Emulab	Named research testbeds; small-scale tests are not credible
Tripwire	Example integrity tool for detecting configuration-file changes

Lecture takeaways

- Flooding looks like a “perfect crime”: cheap volume, spoofed identity, too many bots to punish, and hubs that can congest the whole Internet.
- A box next to the victim cannot keep up; real defence must be **distributed**, but **economics and reputation-secrecy** block the data, benchmarks, and deployment that research needs.
- Ideal properties are **actually stopping attacks**, covering **many attack styles**, **not punishing legitimate users** (including flash crowds), **few false alarms**, and **cost in line with benefit**.
- The operational pipeline is **prevent (especially anti-spoof near sources) → anomaly-detect → identify sources → react without collateral damage**.
- Everyday hygiene — filters, SYN-flood patches, turning off spare services, quotas, baselines, physical security, Tripwire, redundancy, backups, strong admin passwords — **reduces exposure**; it is **not** a complete DDoS solution, which the instructor says the market has not yet delivered.